



**VIDYA VIKAS INSTITUTE OF
ENGINEERING AND TECHNOLOGY,
MYSORE**

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VTU Previous Question Papers

Electronics & Communication Engineering”

III, V & VI Semester

Dec/Jan-2024

VVIET

Bannur road, Alanahalli, Mysore-570028

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Third Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

Basic Signal Processing

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Show that the vectors $(1, 2, 1)$, $(2, 1, 0)$, $(1, -1, 2)$ form a basis of $\mathbb{R}^3$. (08 Marks)
- b. Apply Gram Schmidt process to the vectors $V_1 = (2, 2, 1)$, $V_2 = (1, 3, 1)$, $V_3 = (1, 2, 2)$ to obtain an orthonormal basis for $V_3(\mathbb{R})$ with the standard inner product. (12 Marks)

OR

- 2 a. Reduce the matrix A to echelon form and also find the rank.

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$
(10 Marks)
- b. Determine the null space of each of the following matrices.
 i) $A = \begin{bmatrix} 2 & 0 \\ -4 & 10 \end{bmatrix}$ ii) $B = \begin{bmatrix} 1 & -7 \\ -3 & 21 \end{bmatrix}$
(10 Marks)

Module-2

- 3 a. Find the eigen values and eigen vectors of the matrix

$$A = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$$
(10 Marks)
- b. Diagonalize the following matrix

$$A = \begin{bmatrix} 1 & 3 & 3 \\ -3 & -5 & -3 \\ 3 & 3 & 1 \end{bmatrix}$$
 Find an invertible matrix P and a diagonal matrix D such that $A = PDP^{-1}$. (10 Marks)

OR

- 4 a. Find the singular value decomposition of the matrix $A = \begin{bmatrix} 2 & 3 \\ 0 & 2 \end{bmatrix}$ (10 Marks)
- b. What is the positive definite matrix? If $A = \begin{bmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{bmatrix}$, show that A is positive definite matrix. (10 Marks)

Module-3

- 5 a. Define systems. Explain the communication system with a suitable block diagram. (06 Marks)
- b. Given the signal $x[n]$ show in Fig Q5(b), sketch the following :
- $x[4 - n]$
 - $x[2n + 1]$.

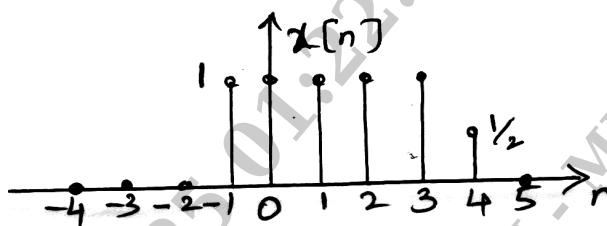


Fig Q5(b)

- c. Explain the following elementary signals

- Unit step signal
- Impulse signal
- Ramp function
- Sinusoidal function.

OR

- 6 a. Explain with an example :
- Amplitude scaling
 - Time scaling
 - Time shifting
 - Precedence rule.
- b. Verify the following system for linearity, time invariance, memoryless, stability and causality.
- $y(n) = nx(n)$
 - $y(n) = 2x[2^n]$.

Module-4

- 7 a. What do you mean by impulse response of an LTI system? Starting from fundamental, deduce the equation for the response of an LTI system, if the input sequences $x[n]$ and the impulse response $h[n]$ are given. (08 Marks)
- b. Find the discrete time convolution sum given below
 $y(n) = \beta^n u(n) * \alpha^n u(n) : |\beta| < 1 \text{ \& } |\alpha| < 1$ (06 Marks)
- c. With suitable diagram, explain the cascade connection and parallel connection of systems. (06 Marks)

OR

- 8 a. A LTI system has an impulse response $h(n) = \begin{cases} 1; n = \pm 1 \\ 2; n = 0 \\ 0; \text{otherwise} \end{cases}$.

Determine the output of this system in response to the input.

$$x(n) = \begin{cases} 2; n = 0 \\ 3; n = 1 \\ -2; n = 2 \\ 0; \text{otherwise} \end{cases}$$

- b. Explain the following properties of system in terms of impulse response

i) Memoryless ii) Causal iii) Stable.

(06 Marks)

- c. Consider the interconnection of four LTI systems as depicted in Fig Q8(c).

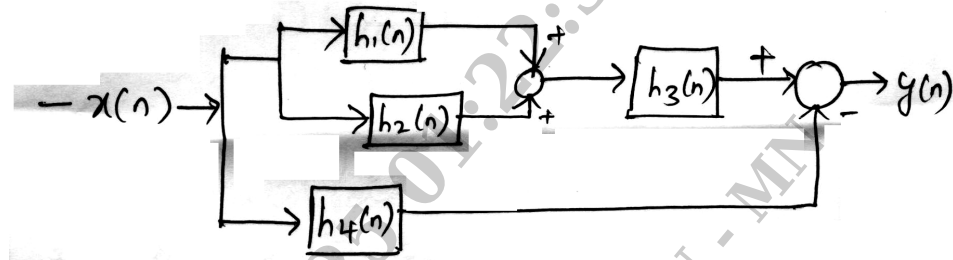


Fig Q8(c)

The impulse response of the systems are

$h_1(n) = U(n)$; $h_2(n) = U(n+2) - U(n)$; $h_3(n) = \delta(n-2)$, $h_4(n) = \alpha^n U(n)$. Find impulse response $h(n)$ of the overall system. (06 Marks)

Module-5

- 9 a. State and prove :

i) Time reversal

ii) Differentiation in Z-domain property of Z-transform.

(08 Marks)

- b. Use partial fraction expansion to find the inverse Z-transform of

$$x(z) = \frac{1+2z^{-1}+z^{-2}}{1-\frac{3}{2}z^{-1}+\frac{1}{2}z^{-2}} \text{ with ROC ; } |z| > 1$$

(12 Marks)

OR

- 10 a. List the properties of z-transform.

(06 Marks)

- b. A causal system has input $x(n]$ and output $y(n]$. find the impulse response of system if

$$x(n) = \delta(n) + \frac{1}{4}\delta(n-1) - \frac{1}{8}\delta(n-2)$$

$$y(n) = \delta(n) - \frac{3}{4}\delta(n-1)$$

(08 Marks)

- c. Find the Z-transform of the signal $x(n) = a^n U(n)$. Indicate the ROC and location of poles and zeros of $x(z)$ in the Z plane. (06 Marks)

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Fourth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

Digital Signal Processing

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Describe the process of frequency domain sampling and reconstruction of discrete time signal. (10 Marks)
- b. Find the 4-point DFT of the sequence $x(n) = \{0, 1, 2, 3\}$ using matrix method and verify the answer by taking the 4 point IDFT of the result. (10 Marks)

OR

- 2 a. State and prove the following properties:
 - i) Circular time shift of a sequence
 - ii) Parseval's theorem. (10 Marks)
- b. Compute the circular convolution of the following sequences using DFT and IDFT method $x_1(n) = \{1, 2, 3, 4\}$ and $x_2(n) = \{4, 3, 2, 1\}$. (10 Marks)

Module-2

- 3 a. Find the output $y(n)$ of a filter whose impulse response $h(n) = \{3, 2, 1, 1\}$ and input $x(n) = \{1, 2, 3, 3, 2, 1, -1, -2, -3, 5, 6, -1, 2, 0, 2, 1\}$ using overlap add method by assuming the 7-point circular convolution. (12 Marks)
- b. Given $x(n) = \{1, 2, 3, 4\}$, find $y(n)$, if $y(k) = x((k-2))_4$. (04 Marks)
- c. Write a program to find the circular convolution of two sequences. (04 Marks)

OR

- 4 a. Compute the 8-point DFT of the sequence $x(n) = \{2, 2, 2, -1, -1, -1, -2, 1\}$ using decimation in time-FFT algorithm. (12 Marks)
- b. Prove the periodicity and symmetry properties of twiddle factor. (08 Marks)

Module-3

- 5 a. A low pass filter is to be designed with the following desired frequency response

$$H_d(e^{j\omega}) = \begin{cases} e^{-j3\omega/4} & \text{for } -3\pi/4 \leq \omega \leq 3\pi/4 \\ 0 & \text{Otherwise} \end{cases}$$

Determine $H(e^{j\omega})$ for $M = 7$ using Hamming window. Also write a program to design low pass FIR filter to meet the specifications. (12 Marks)
- b. Explain the following:
 - i) Rectangular window
 - ii) Hamming window
 - iii) Hanning window. (08 Marks)

OR

- 6 a. A FIR filter is given by

$$y(n) = x(n) + \frac{2}{5}x(n-1) + \frac{3}{4}x(n-2) + \frac{1}{3}x(n-3)$$

 Draw the lattice structure. (10 Marks)
- b. The frequency response of an FIR filter is given by
 $H(W) = e^{-j3W} (1 + 1.8 \cos 3W + 1.2 \cos 2W + 0.5 \cos W)$. Determine the coefficients of the impulse response $h(n)$ of the FIR filter. (06 Marks)
- c. List the advantages of FIR filter. (04 Marks)

Module-4

- 7 a. Realize the following digital filter using direct form I and direct form II:

$$H(Z) = \frac{0.7 + 1.4Z^{-1} + 0.74Z^{-2} + 0.5Z^{-3}}{1 + 1.3Z^{-1} + 0.5Z^{-2} + 0.7Z^{-3} + 0.3Z^{-4}}$$

 Also write a program to design IIR Butterworth low pass filter to meet the specifications. (10 Marks)
- b. Assuming that $T = 2$ sec is bilinear transformation and given the following points :
 i) $S = -1 + j$ on the left half of s-plane
 ii) $S = 1 - j$ on the right half of s place
 iii) $S = j$ on the positive jw on the s-plane
 iv) $S = -j$ on the negative jw on the s-plane.
 Convert each of these points in s-plane to z-plane and verify the mapping properties. (10 Marks)

OR

- 8 a. Using bilinear transformation, design a digital low pass Butterworth filter with the following specifications: sampling frequency: 8000 Hz, 3 dB attenuation at 1.5 kHz, 10 dB stop band attenuation at 3 kHz. (12 Marks)
- b. Discuss the general procedure for IIR filter design using Bilinear transformation. (08 Marks)

Module-5

- 9 a. Discuss briefly :
 i) Multiplier and Accumulator
 ii) Shifters
 iii) Address generators in digital signal processor hardware units (10 Marks)
- b. Explain digital signal processors using Harvard architecture. (10 Marks)

OR

- 10 a. Implement the following IIR filter using Q-15 fixed point format $y(n) = 2x(n) + 0.5y(n-1)$ maximum input is $x_{\max} = (0.0100...0)_2$ in Q-15 format. (10 Marks)
- b. Explain the basic architecture of TMS320CS54X used in fixed point digital signal processor. (10 Marks)

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Fourth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

Circuits and Controls

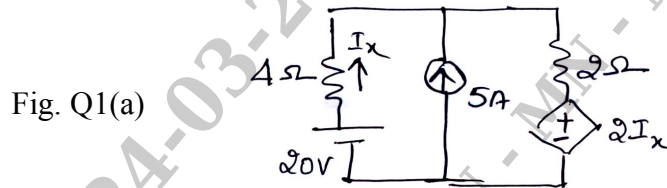
Time: 3 hrs.

Max. Marks: 100

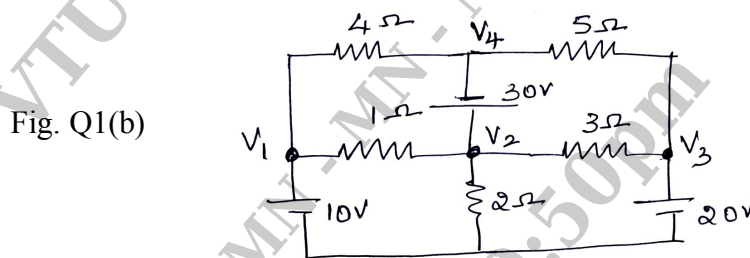
Note : Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

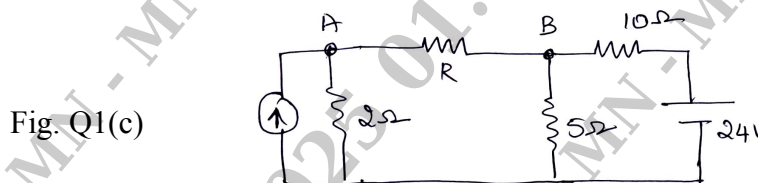
- 1 a. State superposition theorem and using superposition theorem, determine I_x for the circuit in Fig. Q1(a). (08 Marks)



- b. Determine the node voltages for the network shown in Fig. Q1(b) using nodal analysis. (06 Marks)

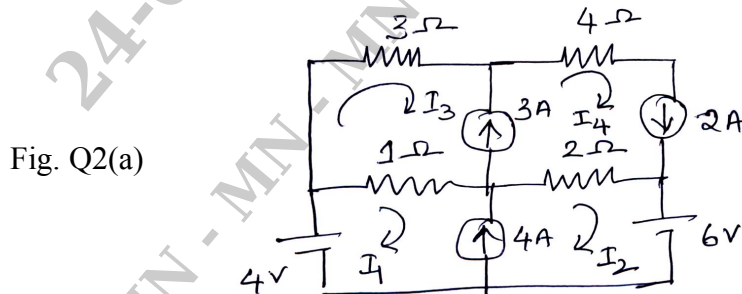


- c. Find the value of R such that maximum power transfer can take place for the circuit shown in Fig. Q1(c). Also find the maximum power delivered. (06 Marks)



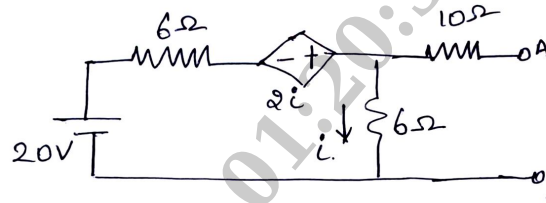
OR

- 2 a. Determine the mesh current for the network shown in the Fig. Q2(a) using mesh analysis method. (06 Marks)



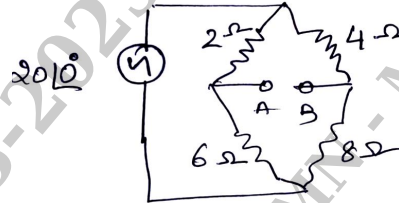
- b. State Thevenin's theorem and determine the Thevenin's equivalent network between A & B for the network shown in Fig. Q2(b). (08 Marks)

Fig. Q2(b)



- c. Obtain Norton's equivalent circuit for the network shown in Fig. Q2(c). (06 Marks)

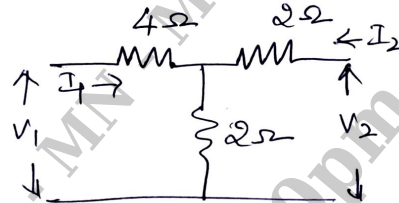
Fig. Q2(c)



Module-2

- 3 a. Determine 'Z' parameter for the network shown in Fig. Q3(a). (06 Marks)

Fig. Q3(a)



- b. Given $I(s) = \frac{2s+5}{(s+1)(s+2)}$. Determine the initial and final value of $i(t)$ using initial and final value theorem. (08 Marks)

- c. For the given Z parameter equation, find ABCD parameter.

$$V_1 = 6I_1 + 20I_2$$

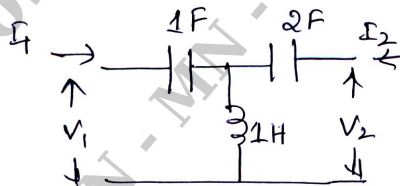
$$V_2 = 20I_1 + 40I_2$$

(06 Marks)

OR

- 4 a. Determine Z parameter and 'h' parameter for the network shown in Fig. Q4(a). (10 Marks)

Fig. Q4(a)



- b. Determine 'Z' and 'Y' parameter for the network shown in Fig. Q4(b). (10 Marks)

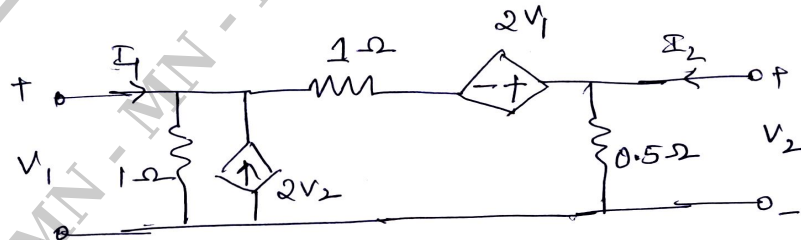
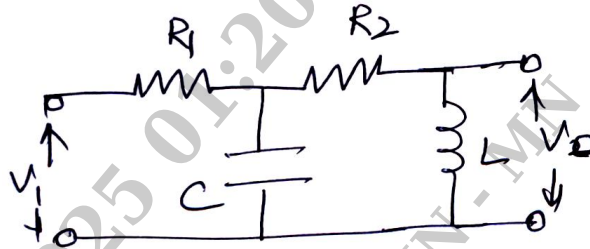


Fig. Q4(b)

Module-3

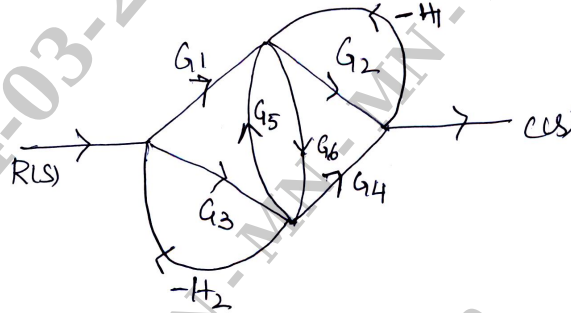
- 5 a. Define Open loop system and closed loop system and mention all the merits and demerits of open loop and closed loop system. (06 Marks)
- b. Write differential equation for the network given in Fig. Q5(b). (06 Marks)

Fig. Q5(b)



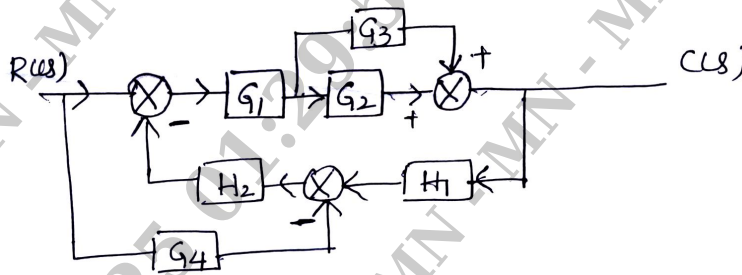
- c. Obtain transfer function for the given signal flow in Fig. Q5(c). (08 Marks)

Fig. Q5(c)

**OR**

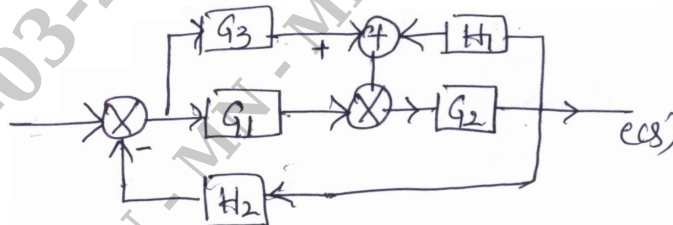
- 6 a. Simplify the given block as shown in Fig. Q6(a) using block reduction method and find its transfer function. (10 Marks)

Fig. Q6(a)



- b. Draw signal flow graph for the given block as shown in Fig. Q6(b) and obtain transfer function using Mason's Gain formula. (10 Marks)

Fig. Q6(b)

**Module-4**

- 7 a. Obtain the time response of a second order system subjected to unit step input for under damped oscillation. (08 Marks)
- b. Check the stability of the given characteristic equation using Routh Hurwitz criteria. The characteristic equation is $F(s)$.
 $F(s) = s^6 + 2s^5 + 8s^4 + 12s^3 + 20s^2 + 16s + 16$. Also find roots on RHS, LHS and on imaginary axis. (06 Marks)

- c. A second order system is given by

$$\frac{C(s)}{R(s)} = \frac{25}{s^2 + 6s + 25}. \text{ Find Rise time, Settling time, Peak over shoot and Peak time.}$$

(06 Marks)

OR

- 8 a. Using RH criteria find K_{mar} and 'w' at K_{mar} given characteristic equation $s^4 + 22s^3 + 10s^2 + s + k = 0$.

(06 Marks)

- b. For a system having $G(s) = \frac{15}{(s+1)(s+3)}$ and $H(s) = 1$.

Find W_n , damping ratio ζ , W_d and θ . Also find $C(t)$, T_p , M_p and T_s (for 2% variance).

(08 Marks)

- c. For the given characteristics equation $F(s) = s^2 + 4s + 4$. Using RH criteria determine the stability of the system and determine roots lying between $S = 0$ and $S = -1$.

(06 Marks)

Module-5

- 9 a. Obtain the Root locus for the given transfer function

$$G(s)H(s) = \frac{K}{s(s+3)(s^2 + 3s + 11.25)}.$$

Find the range of 'K' for stability.

(10 Marks)

- b. Find the state model equation for the differential equation given below :

$$\frac{dy^4}{dt^4} + 10\frac{d^3y}{dt^3} + 10\frac{d^2y}{dt^2} + \frac{dy}{dt} + y(t) = 10x(t).$$

(10 Marks)

OR

- 10 a. Find the state transition matrix for

$$A = \begin{bmatrix} 0 & -1 \\ 2 & -3 \end{bmatrix}.$$

(08 Marks)

- b. Explain properties of state transition matrix.

(06 Marks)

- c. A unity feedback control system has $G(s) = \frac{80}{s(s+2)(s+20)}$.

Draw a Bode plot and find W_{gc} and W_{pc} . Comment on stability.

(06 Marks)

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CBCS SCHEME

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21EC52

Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Computer Organization and ARM Microcontrollers

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain the pipelining and superscalar operation. (05 Marks)
- b. Briefly explain the different key parameters that affects the processor performance. (05 Marks)
- c. With a neat diagram, explain basic operational concept of computer. (10 Marks)

OR

- 2 a. Mention the difference between Big-endian and Little-endian assignments. (05 Marks)
- b. Explain the different types of addressing modes. (10 Marks)
- c. With diagram explain the Interrupt Hardware. (05 Marks)

Module-2

- 3 a. With diagram explain the internal organization of a 2M×8 dynamic memory chip. (10 Marks)
- b. Explain the operations of synchronous DRAM. (10 Marks)

OR

- 4 a. Explain the different types of memories. (10 Marks)
- b. With neat diagram explain the multiple bus organization. (10 Marks)

Module-3

- 5 a. Explain the architecture of ARM core dataflow model. (10 Marks)
- b. With neat diagram explain the ARM based embedded system. (10 Marks)

OR

- 6 a. Explain the ARM condition flag register. (05 Marks)
- b. What is pipeline? Explain the 3-stage ARM pipeline. (05 Marks)
- c. Explain the architecture of ARM processor. (10 Marks)

Module-4

- 7 a. With example explain the following instructions:
i) ADC ii) EOR iii) SWI iv) UMULL v) SBC (10 Marks)
- b. With neat diagram explain the ARM stack operation. (10 Marks)

OR

- 8 a. Explain the ARM Registers and also explain each. (10 Marks)
- b. Explain the following ARMVSE extension instructions:
i) CLZ ii) QADD iii) QSUB iv) SMLAxy v) QDADD (10 Marks)

Module-5

- 9 a. Explain the following THUMB instructions:
i) TST ii) ROR iii) BX iv) BKPT v) ASR (10 Marks)
- b. Discuss the ARM support basic C data types. (10 Marks)

OR

- 10 a. Explain the ARM function calls and loop operations. (10 Marks)
- b. With example explain the single register and multiple register load store ARM instructions. (10 Marks)

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Important Note : 1. On completing your answers, compulsorily draw diagonal cross lines on the remaining blank pages.
2. Any revealing of identification, appeal to evaluator and /or equations written eg, 42+8 = 50, will be treated as malpractice.

CBCS SCHEME

USN

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21EC54

Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Electromagnetic Waves

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. If two position vectors $\vec{A} = -2\vec{a}_x - 5\vec{a}_y - 4\vec{a}_z$ and $\vec{B} = 2\vec{a}_x + 3\vec{a}_y + 5\vec{a}_z$ then find,
i) $\vec{AB}$ ii) $\vec{a}_A, \vec{a}_B$ iii) $\vec{a}_{AB}$ iv) Unit vector from C to A where C is (3, 5, 8) (06 Marks)
- b. Ten identical charges each of 500 μC are spaced equally around a circle of radius 2 m. Find the force on a charge of $-20 \mu\text{C}$ located on the axis, 2m from the plane of the circle. (07 Marks)
- c. Define Electric Field Intensity. Derive expression for electric field intensity due to 'n' number of charges. (06 Marks)

OR

- 2 a. Given the two points A(2, 3, -1) and B(4, 25, 120). Find spherical coordinates of A and Cartesian coordinates of B. (06 Marks)
- b. Derive an expression for electric field intensity due to infinite line charge. (07 Marks)
- c. Find electric field $\vec{E}$ at origin, if the following charge distribution are present in free space :
i) Point charge of 21nC at P(2, 0, 6)
ii) Uniform line charge of infinite length with charge density $\rho_l = 3 \text{ nC/m}$ at $x = 2, y = 3$.
iii) Uniform surface charge of density 0.2 nC/m^2 at $x = 2$. (07 Marks)

Module-2

- 3 a. A charge is uniformly distributed over a spherical surface of radius 'a'. Determine electric field intensity at all the places, use Gauss law. (07 Marks)
- b. Evaluate both sides of divergence theorem for the field $\vec{D} = 2xy\vec{a}_x + x^2\vec{a}_y \text{ C/m}^2$, for a rectangular parallel piped formed by the planes $x = 0$ and $x = 1$; $y = 0$ and $y = 2$; $z = 0$ and $z = 3$. (08 Marks)
- c. Show that electric field intensity is negative potential gradient. (05 Marks)

OR

- 4 a. The flux density $\vec{D} = \frac{r}{3} \vec{a}_r \text{ nC/m}^2$ in free space :
i) Find $\vec{E}$ at $r = 0.2 \text{ m}$
ii) Find the total electric flux leaving the sphere of $r = 0.2 \text{ m}$.
iii) Find the total charge within the sphere of $r = 0.3 \text{ m}$ (07 Marks)
- b. State and prove Gauss divergence theorem. (07 Marks)
- c. Derive an expression for continuously equation. (06 Marks)

Module-3

- 5 a. Determine whether or not the following potential fields satisfy the Laplace's equation:
 i) $V = x^2 - y^2 + z^2$ ii) $V = r \cos\phi + z$ iii) $V = r \cos\theta + \phi$ (06 Marks)
- b. Evaluate both sides of Stoke's theorem for the field $\vec{H} = 6xy\vec{a}_x - 3y^2\vec{a}_y$ A/m, for rectangular path around the region, $2 \leq x \leq 5$, $-1 \leq y \leq 1$, $z = 0$. (08 Marks)
- c. Explain the concept of magnetic potential. (06 Marks)

OR

- 6 a. State and prove uniqueness theorem. (06 Marks)
- b. Two plates of parallel plate capacitor are separated by distance 'd' and maintained at potential zero and V_0 respectively. Determine,
 i) Potential at any position between the plates
 ii) Surface charge density on the plates
 iii) Capacitance between the plates. (08 Marks)
- c. Find the magnetic flux density at the centre 'O' of a square of sides equal to 5 m and carrying 10 A of current. (06 Marks)

Module-4

- 7 a. Derive an expression for force on a differential current element and find force experienced by conductor of 6 m long, lies along z-direction with a current of 2A in $\vec{a}_z$ direction, if $\vec{B} = 0.08\vec{a}_x$ T. (07 Marks)
- b. Explain magnetization and permeability. (07 Marks)
- c. Derive boundary conditions at the interface of two magnetic materials. (06 Marks)

OR

- 8 a. A point charge of $Q = -1.2$ C has velocity $\vec{v} = (5\vec{a}_x + 2\vec{a}_y - 3\vec{a}_z)$ m/s. Find the magnitude of force exerted on the charge if,
 i) $\vec{E} = -18\vec{a}_x + 5\vec{a}_y - 10\vec{a}_z$ V/m
 ii) $\vec{B} = -4\vec{a}_x + 4\vec{a}_y + 3\vec{a}_z$ T
 iii) Both are present simultaneously. (07 Marks)
- b. Find the magnetization in a magnetic material where:
 i) $\mu = 1.8 \times 10^5$ H/m and $M = 120$ A/m
 ii) $\mu = 22$, there are 8.3×10^{28} atoms/m³ each atom has a dipole moment of 4.5×10^{-27} A/m² and
 iii) $B = 300$ μ T and $X_m = 15$ (07 Marks)
- c. State and explain Faraday's law of electromagnetic induction. Hence obtain Maxwell's equation in point form and integral form. (06 Marks)

Module-5

- 9 a. Derive Maxwell's equations for time varying fields, represent them in point form and integral form. (08 Marks)
- b. Obtain relationship between $\vec{E}$ and $\vec{H}$ in free space. (06 Marks)

- c. In free space $\vec{E} = 50 \cos(\omega t - \beta z) \vec{a}_x$ V/m. Find the average power crossing a circular area of radius 2.5m in the plane $z = 0$. (06 Marks)

OR

- 10 a. Given $\vec{E} = E_m \sin(\omega t - \beta z) \vec{a}_y$ in free space, find $\vec{D}$, $\vec{B}$ and $\vec{H}$. Sketch $\vec{E}$ and $\vec{H}$ at $t = 0$. (08 Marks)
- b. Explain wave propagation in good conductor with relevant equations. (08 Marks)
- c. Wet marshy soil is characterized by $\sigma = 10^{-2}$ S/m, $\epsilon_r = 15$ and $\mu_r = 1$. At 1 MHz whether soil be considered as conductor or dielectric. (04 Marks)

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21EC/EI61

Sixth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

Technological Innovation Management and Entrepreneurship

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define Management and explain the functions of management. (08 Marks)
- b. Is management a science or art? Explain. (06 Marks)
- c. Explain the various roles of manager. (06 Marks)

OR

- 2 a. Explain the different levels of management along with skills required. (07 Marks)
- b. Distinguish between Management and Administration. (05 Marks)
- c. Explain the various steps involved in decision making process. (08 Marks)

Module-2

- 3 a. Briefly explain the principles of organization. (10 Marks)
- b. What is Departmentalisation? Explain the advantages and disadvantages of committees in an organization. (10 Marks)

OR

- 4 a. Explain Maslow's hierarchy need theory. (10 Marks)
- b. Explain the various steps involved in selection process. (10 Marks)

Module-3

- 5 a. What is the meaning of social responsibility of business? Explain Social responsibility of business towards different groups. (10 Marks)
- b. What is a social audit? Explain its benefits and limitations. (10 Marks)

OR

- 6 a. Write a short note on behavioural approach of leadership. (04 Marks)
- b. Explain any four essentials of an effective control system. (08 Marks)
- c. What is corporate governance? List the benefits of Good Corporate Governance. (08 Marks)

Module-4

- 7 a. Briefly explain any four characteristics of a successful entrepreneurship. (10 Marks)
- b. Explain the strategies in capacity building for entrepreneurship. (10 Marks)

OR

- 8 a. Discuss about the business opportunities in India. (10 Marks)
b. Write a short note on Mobility of Entrepreneurs. (10 Marks)

Module-5

- 9 a. Explain the formalities for starting business enterprise. (10 Marks)
b. Discuss the reasons behind the failure of certain business plan. (10 Marks)

OR

- 10 a. Briefly explain about NGO's (Non governmental organizations). (10 Marks)
b. Write a short note on:
(i) TCO's (Technical Consultancy Organizations)
(ii) SIDBI (Small Industries Development Bank of India). (10 Marks)

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Sixth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

Microwave Theory and Antennas

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

1. a. Derive Transmission line Equations in voltage and current forms. (08 Marks)
 b. List the characteristics of Smith Chart. (07 Marks)
 c. A certain transmission line has a characteristic impedance of $75 + j0.01 \Omega$ and is terminated in a load impedance of $70 + j50 \Omega$, compute
 (i) Reflection co-efficient
 (ii) Transmission co-efficient. (05 Marks)

OR

2. a. With neat diagram, explain the typical Microwave system. (07 Marks)
 b. A transmission line has the following parameters, $R = 2\Omega/m$, $G = 0.5 \text{ mmho/m}$, $f = 1 \text{ GHz}$, $L = 8 \text{ nH/m}$, $C = 0.23 \text{ PF}$.
 Calculate (i) Characteristic impedance (ii) Propagation constant (05 Marks)
 c. With the help of a functional block diagram, explain construction and modes of working of a GUNN Diode. (08 Marks)

Module-2

3. a. Write the S-matrix representation for Mutliport network. (07 Marks)
 b. With a neat diagram, explain the working of precession type variable attenuator. (08 Marks)
 c. A 20 mW signal is fed into one of the collinear Port 1 of a lossless H-plane T-junction. Calculate the power delivered through each port when other ports are terminated in Matched load. (05 Marks)

OR

4. a. State and explain the properties of S-matrix. (07 Marks)
 b. What is Magic-tee? Derive its scattering matrix. (07 Marks)
 c. In an H-plane T-junction, compute power delivered to the loads of 40 ohm and 60 ohm connected to arms-1 and 2 when a 10 mW power is delivered to the matched Port 3. (06 Marks)

Module-3

5. a. Discuss briefly Micro-strip lines and its losses and also derive the expression for Quality factor. (08 Marks)
 b. Define the following terms with respect to antennas:
 (i) Radiation Intensity
 (ii) Beam area.
 (iii) Directivity
 (iv) Beam efficiency. (08 Marks)
 c. A radio link has a 15 W transmitter connected to an antenna of 2.5 m^2 effective aperture at 5 GHz. The receiving antenna has an effective aperture of 0.5 m^2 and is located at 15 km line of sight distance from the transmitting antenna. Assume lossless antennas. Find the power delivered to the receiver. (04 Marks)

OR

- 6 a. A lossless parallel strip line has a conducting strip width W . The substrate dielectric separating the two conducting strips has a relative dielectric constant ϵ_{rd} of 6 and a thickness d of 4 mm. Calculate
- The required width W of the conducting strip in order to have a characteristic impedance of 50Ω .
 - The strip line capacitance.
 - The strip line inductance.
 - The phase velocity of the wave in the parallel strip lines. (08 Marks)
- b. Show that maximum effective aperture of a $\lambda/2$ dipole antenna is $0.13 \lambda^2$. (06 Marks)
- c. Find the Directivity of the following using Exact method,
- $U = U_m \sin^2 \theta \sin^3 \phi$
 - $U = U_m \cos^4 \theta \sin^2 \phi$ where $0 \leq \theta \leq \frac{\pi}{2}$ and $0 \leq \phi \leq 2\pi$. (06 Marks)

Module-4

- 7 a. Derive the expression and draw the field pattern for two isotropic point sources of the same Amplitude and same phase. (08 Marks)
- b. Derive the expression for radiation resistance of short electric dipole. (08 Marks)
- c. Find the length of an Elementary dipole having a radiation resistance of 5Ω at a frequency of 5 MHz. (04 Marks)

OR

- 8 a. Derive array factor expression in case of n isotropic point sources of equal Amplitude and spacing. (07 Marks)
- b. Starting from Electric and Magnetic field potential, obtain the far field components for a short dipole. (10 Marks)
- c. For a short dipole $\frac{\lambda}{15}$ long find the radiation resistance. (03 Marks)

Module-5

- 9 a. Obtain the expression for radiation resistance of small loop antenna. (07 Marks)
- b. Briefly explain helical antenna with its helical geometry. (05 Marks)
- c. Find the length L , H-plane aperture and flare angle θ_E and θ_H of pyramidal horn for which E-plane aperture is 10λ Horn is fed by a rectangular waveguide with TE_{10} mode. Assume $\delta = 0.2\lambda$ in E-plane and 0.375λ in H-plane. Also find beam widths and directivity. (08 Marks)

OR

- 10 a. Derive the expression for strength E_ϕ and H_θ in case of small loop Antenna. (08 Marks)
- b. Explain different types of Horn Antenna. (06 Marks)
- c. Design Yagi-Uda antenna of six elements to provide a gain of 12 dB, if the operating frequency is 200 MHz. (06 Marks)

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21EC63

Sixth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 VLSI Design and Testing

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Realize the CMOS compound gate for the following functions:
(i) $Y = \overline{ABC + D}$ (ii) $Y = \overline{A(B + C) + DE}$ (08 Marks)
b. Implement a positive edge triggered D flip flop using transmission gate and write the necessary timing diagram. (08 Marks)
c. Analyze the working of tristate inverter. (04 Marks)

OR

- 2 a. Draw the circuit diagram of CMOS inverter and derive its transfer characteristics by graphical method. (06 Marks)
b. Derive the equation for drain current of a MOSFET in non-saturated and saturated region of operation. (10 Marks)
c. Explain the following non-ideal effects of MOSFETs:
(i) Channel length modulation (ii) Mobility degradation. (04 Marks)

Module-2

- 3 a. With neat diagrams, explain the complete CMOS n-well fabrication process. (10 Marks)
b. Draw the layout diagram for the following function and also estimate the area.
 $Y = \overline{(A + B + C)D}$ (10 Marks)

OR

- 4 a. Using Elmore delay model estimate the t_{pdf} and t_{pdr} of a 3-input NAND gate if the output is loaded with 'h' identical gates. (08 Marks)
b. Find the logical effort and parasitic delay of (i) 2 input NOR gate and (ii) 3-input NAND gate. (06 Marks)
c. Construct necessary equivalent circuit for the computation of t_{pdf} of an inverter driving another inverter using RC delay model. (06 Marks)

Module-3

- 5 a. With necessary circuit diagrams explain the operation of (i) 4 transistor DRAM and (ii) 3 transistor DRAM cells. (10 Marks)
b. Explain the operation of full CMOS SRAM cell with necessary circuit topology. (06 Marks)
c. Explain the hysteresis characteristics of ferroelectric capacitor with necessary diagram. (04 Marks)

OR

- 6 a. Explain the operation of 4×4 NOR based ROM array with necessary circuit diagram. (08 Marks)
b. With necessary circuit diagram and bias conditions, explain the operation of NOR flash memory. (06 Marks)
c. Explain binary tree based column decoder design with necessary diagram. (06 Marks)

Module-4

- 7 a. Differentiate between fault and failure with an example. Explain different types of stuck at fault with example. (06 Marks)
 b. Explain feedback bridging fault with an example. (06 Marks)
 c. For the circuit shown in Fig.Q7(c), using Boolean difference (i) detect s-a-0 and s-a-1 at x_1 , (ii) determine partial Boolean difference for $x_2 - 1 - n - p - F$.

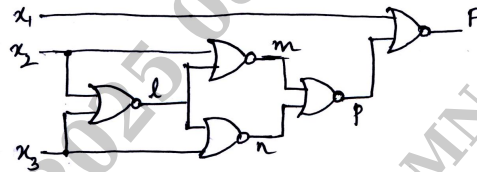


Fig.Q7(c)

(08 Marks)

OR

- 8 a. What is fault diagnosis? Explain delay fault detection with an example. (08 Marks)
 b. Find the test pattern for line 6 s-a-0 for the circuit shown in Fig.Q8(b) using D algorithm.

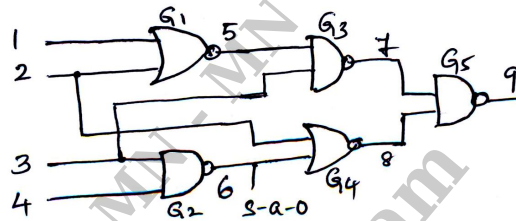


Fig.Q8(b)

(12 Marks)

Module-5

- 9 a. For the state table shown in Table Q9(a), find
 (i) Response for 010 sequence, (ii) Homing tree, (iii) Distinguishing tree.

Present State	Input	
	$x = 0$	$x = 1$
A	B, 0	D, 0
B	A, 0	B, 0
C	D, 1	A, 0
D	D, 1	C, 0

Table Q9(a)

(10 Marks)

- b. Write a note on functional fault model to detect faults in sequential circuits. (05 Marks)
 c. Explain the process of testing sequential circuit as iterative combinational circuits. (05 Marks)

OR

- 10 a. Define the terms controllability and observability with an example. (05 Marks)
 b. With a neat logic diagram, explain clocked hazard free latches used in LSSD technique. (08 Marks)
 c. Explain partial scan technique using system clock with necessary diagram. (07 Marks)

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21EC642

Sixth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Cryptography

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain Euclidean algorithm to find the GCD of two numbers with the help of an example where $a = 234$, $b = 42$. (10 Marks)
- b. What is modulus and congruent modulo n . State the properties of congruences with the help of an example for each. (10 Marks)

OR

- 2 a. Explain division algorithm with the help of number line. And also state the properties of divisibility for integers. (10 Marks)
- b. Explain the procedure used for finding the multiplicative inverse in $GF(P)$ perform arithmetic operations in $FG(7)$. (10 Marks)

Module-2

- 3 a. With the help of neat block diagram, explain the model for network security. (10 Marks)
- b. Explain the procedure involved in encrypting a message using play-fair cipher. A message to be sent at an wireless station in play-fair cipher is "cryptography" using the key "MONARCHY" encrypt the message. (10 Marks)

OR

- 4 a. Explain the procedure involved in encrypting a message using hill cipher for $m = 3$ encrypt the message "herbre" using the hill cipher with the key $\begin{bmatrix} 3 & 7 \\ 5 & 12 \end{bmatrix}$. Find whether decryption is possible or not. Show your calculations and result. (10 Marks)
- b. Explain passive and active attacks in network security. (10 Marks)

Module-3

- 5 a. With the help of neat block diagram, explain the salient features of DES encryption algorithm. (10 Marks)
- b. Explain Euler's theorem with example as $a = 2$, $n = 11$. (10 Marks)

OR

- 6 a. Given the plaintext $\{000102030405060708090A0B0C0D0E0F\}$ and the key $\{01010101010101010101010101010101\}$
 - i) Show the original contents of state, displayed as a 4×4 matrix
 - ii) Show the value of state after initial Addround key
 - iii) Show the value of state after shift rowsMake use of AES algorithm. (10 Marks)
- b. Explain the choice of the parameters used in the design of traditional block cipher/Feistel cipher. (10 Marks)

Module-4

- 7 a. Explain the applications and requirements of public –key cryptosystems. (10 Marks)
b. Assuming $p = 7$ and $q = 17$, find the public key and private key. Perform encryption and decryption for plain text message block $M = 6$. (10 Marks)

OR

- 8 a. Show that in Diffie – Hellman key exchange algorithm, the keys generated at sender side and receiver side are same. Assuming $q = 23$ and $\alpha = 5$, users A and B select their private keys $X_A = 6$ and $X_B = 15$ compute their public key Y_A and Y_B and shared secret key K. (10 Marks)
b. Describe the Elliptic curve cryptography. (10 Marks)

Module-5

- 9 a. Explain LFSR and how the shift register sequences are used in cryptography. (10 Marks)
b. Write a note on design and analysis of stream cipher. (10 Marks)

OR

- 10 a. With a neat diagram, explain generalized Geffe generator. (10 Marks)
b. Write short notes on :
i) A5 to encrypt GSM
ii) NANOTEQ and RAMBUTAN. (10 Marks)

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21EC71

Seventh Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Advanced VLSI

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain ASIC design flow with neat flow chart. (10 Marks)
- b. Explain Booth Encoding multiplier with an example. (10 Marks)

OR

- 2 a. Describe different cell compilers and I/O cells. (10 Marks)
- b. With a neat diagram, explain the operation of conditional sum adder. Mention its advantages and disadvantages. (10 Marks)

Module-2

- 3 a. Explain the concept of measurement of delay in Floorplanning. (10 Marks)
- b. Explain the following : (10 Marks)
 - i) Power planning
 - ii) Clock planning

OR

- 4 a. Write an algorithm for iterative placement improvement method and explain briefly. (10 Marks)
- c. Explain the goals and objectives of global routing in detail. (10 Marks)

Module-3

- 5 a. Explain factors in randomizing the stimulus to a design. (10 Marks)
- b. Draw the diagram of layered test bench of system verilog and describe the function of each layer. (10 Marks)

OR

- 6 a. Describe the various array methods with an example. (08 Marks)
- b. Describe type def and enumerated data types with example. (06 Marks)
- c. Explain constant and strings in system verilog with example. (06 Marks)

Module-4

- 7 a. Explain Tasks, functions and void function in system verilog. (06 Marks)
- b. How time values are specified in system verilog, describe with example. (06 Marks)
- c. Explain automatic storage and variable initialization with system verilog program example. (08 Marks)

OR

- 8 a. Describe the communication between the test bench and DUT with suitable diagram and system verilog program. (10 Marks)
- b. Explain different types of system verilog assertions with example. (10 Marks)

Module-5

- 9 a. Explain common randomization problems in brief. (10 Marks)
b. Write any 4 Random Number functions with an example. (04 Marks)
c. Explain Pseudorandom Number Generators (PRNG) in system verilog. (06 Marks)

OR

- 10 a. What is coverage? Explain coverage types in system verilog. (10 Marks)
b. Describe various coverage options with an example. (10 Marks)

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21EC72

Seventh Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Optical and Wireless Communication

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Briefly explain with a neat figure the propagation mechanism of meridional rays in an ideal step index optical waveguide. (08 Marks)
- b. Define the term attenuation in optical fibers. Explain the different attenuation mechanisms in optical fibers. (12 Marks)

OR

- 2 a. Define Dispersion. Briefly explain intermodal and intramodal dispersion effects in optical waveguide. (10 Marks)
- b. With neat figures, discuss the structure of single mode and multimode step-index and graded index optical fibers. (06 Marks)
- c. A multimode fiber has a core refractive index of 1.480 and a core cladding index difference of 2.0 percent. Find the numerical aperture and critical angle at the core cladding interface. (04 Marks)

Module-2

- 3 a. What are the characteristic requirements of an optical source? With the help of neat diagram, explain the constructional features and emission pattern of surface emitting LED. (10 Marks)
- b. Define optical isolator. With a neat figure, explain the design and operation of a polarization independent isolator. (06 Marks)
- c. A given silicon avalanche photodiode has a quantum efficiency of 65 percent at a wavelength of 900 nm. If 0.5 μW of optical power produces a multiplied photocurrent of 10 μA . What is the multiplication M? (04 Marks)

OR

- 4 a. Discuss the operation of pin photodiode with a neat circuit and energy band diagram. (10 Marks)
- b. What is Diffraction gratings? Discuss briefly Diffraction grating techniques. (10 Marks)

Module-3

- 5 a. Explain briefly the different propagation mechanisms that influence the signal propagation in a mobile communication environment. (10 Marks)
- b. A cellular communication service area is covered with 12 clusters having 7 cells in each cluster and 16 channels assigned in each cell. Find the number of channels per cluster and the system capacity. (03 Marks)
- c. Explain how the concept of frequency reuse increases the spectrum efficiency that in turn increases the cellular communication system capacity. (07 Marks)

OR

- 6 a. Briefly discuss the generations of wireless communication network technology. (08 Marks)
- b. Discuss the effects of co-channel interference in wireless communication in reducing the system capacity. (05 Marks)
- c. Discuss the concept of multipath fading in mobile communication system. (07 Marks)

Module-4

- 7 a. With a neat block diagram, explain the operation of basic TDMA link. (10 Marks)
- b. Explain the basic cellular system with necessary block diagram. (10 Marks)

OR

- 8 a. Discuss with a neat figure the call processing in a cellular system for mobile-originated calls. (12 Marks)
- b. List the advantages of CDMA over TDMA and FDMA. (08 Marks)

Module-5

- 9 a. What is Hand off in GSM networks? Explain briefly the different handoff procedure in GSM. (10 Marks)
- b. Explain the functions of data bases HLR and VLR at MSC in GSM network architecture and also explain how it is helpful in location updation in GSM networks. (10 Marks)

OR

- 10 a. Briefly explain the three major subsystems in GSM network architecture with a neat block diagram. (10 Marks)
- b. Explain briefly the following identifiers in GSM system:
- (i) SIM
 - (ii) Mobile system ISDN with frame format
 - (iii) Location Area Identify
- (10 Marks)

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21EC732

Seventh Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Digital Image Processing

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. With neat diagram explain Single image sensor, how it can be used in Sensor Strip and Sensor Array. (08 Marks)
- b. Explain basic concept of Sampling and Quantization with reference to Digital Image. (07 Marks)
- c. Calculate the photon energy for visible light for given wavelength range 400 nm to 750 nm. [Plank's constant, $h = 6.63 \times 10^{-34}$ Js, $C = 3 \times 10^8$ m/s] (05 Marks)

OR

- 2 a. Explain the Brightness Adaption and Discrimination. (07 Marks)
- b. Explain the Neighbour pixel basic relationship in Digital Images with adjacency connectivity, Regions and Boundaries. (08 Marks)
- c. Given two pixels P and Q with coordinate positions (-2, -2) and (3, 4) respectively, calculate the distance measure D_e , D_4 , D_8 . (05 Marks)

Module-2

- 3 a. Define 2-D orthogonal and unitary transform. (06 Marks)
- b. For given orthogonal matrix A and an image u obtain unitary transform.
Given $A = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ $u = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ (08 Marks)
- c. Define the properties of unitary transform. (06 Marks)

OR

- 4 a. Define 2-D DFT and its properties. (06 Marks)
- b. Define cosine transform and its properties. (06 Marks)
- c. Calculate Haar transform for $N = 4$

$$\text{Given Haar function } H_a(z) = \frac{1}{\sqrt{N}} \begin{cases} +2^{p/2} & , \quad \frac{q-1}{2^p} \leq z < \frac{q-0.5}{2^p} \\ -2^{p/2} & , \quad \frac{q-0.5}{2^p} \leq z < \frac{q}{2^p} \\ 0 & , \quad \text{else} \end{cases}$$

$$n = \log_2 N$$

$$p = 0 \text{ to } n-1$$

$$q \text{ range between } 1 \leq q \leq 2^p$$

$$k = 2^p + q - 1$$

$$z = 0, 1/4, 2/4, 3/4$$

(08 Marks)

Module-3

- 5 a. With necessary graph and equation explain
 i) Image Negative
 ii) Power law transformation
 iii) Intensity level slicing (06 Marks)
 b. Compute Histogram equalization for given data:

Table 5(b)

r_k	0	1	2	3	4	5	6	7
n_k	790	1023	850	656	329	245	122	81

for 3 bit image ($L = 8$) of size 64×64 pixels ($MN = 4096$) with intensity distribution shown in Table 5(b). Intensity level are integer in range $[0, L-1] = [0, 7]$ (08 Marks)

- c. With an example for 2-bit image of size 5×5 define the sample mean, sample variance with equation. (06 Marks)

OR

- 6 a. Explain with example fundamentals of Spatial Filtering for spatial correlation and convolution for 1-D and 2-D filter. (08 Marks)
 b. Using 1st order derivative Image Sharpening (the Gradient) define:
 i) Robert's cross gradient operation
 ii) Sobel's operators (for 3×3 region) (06 Marks)
 c. Define smoothing spatial filters with brief note:
 i) Linear Filters
 ii) Order Statistic Filter (06 Marks)

Module-4

- 7 a. With neat block diagram of Homomorphic system, derive Homomorphic filtering approach for Image Enhancement. (08 Marks)
 b. Define sharpening of images in frequency domain using
 i) Ideal High Pass Filter
 ii) Butterworth High Pass Filter
 iii) Gaussian High Pass Filter (06 Marks)
 c. Give Frequency domain filtering necessary steps followed. (06 Marks)

OR

- 8 a. Define pseudo color image processing with intensity slicing and intensity to color transformation. (06 Marks)
 b. Based on Hardware oriented models classify different color model given color conversion for RGB to HIS and vice versa with relevant equation. (08 Marks)
 c. With color fundamentals for primary and secondary colors. (06 Marks)

Module-5

- 9 a. Write brief note on restoration in presence of only noise using
 i) Mean filter ii) Order statistic filter iii) Adaptive filter (08 Marks)
 b. Discuss some of the important noise probability density functions. (06 Marks)
 c. With help of block diagram give details of Degradation / Restoration process. (06 Marks)

OR

- 10 a. In digital images discuss about Inverse Filtering. (06 Marks)
 b. Explain minimum mean square error (Wiener Filter) in Digital Image Processing. (08 Marks)
 c. Discuss periodic noise reduction by frequency domain filtering. (06 Marks)

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Seventh Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

Network Security

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Discuss the four principles of security in detail. (10 Marks)
b. What are the 2 types of security attacks? Explain Passive attacks in detail along with classification. (10 Marks)

OR

- 2 a. Write short notes on virus, worms and cookies. (08 Marks)
b. What is packet spoofing? Mention its three possible cases. (06 Marks)
c. Explain sniffing and phishing attacks. (06 Marks)

Module-2

- 3 a. Explain the parameters that define the session state and connection state in the secure socket layer. (08 Marks)
b. Mention the types of security threats faced when using the web. (08 Marks)
c. With the help of diagram, explain the alert protocol and its operation. (04 Marks)

OR

- 4 a. With suitable diagram, explain how the Message Authentication Code (MAC) is calculated and the generation of pseudorandom function in transport layer security. (10 Marks)
b. Explain the connection initiation and closure of HTTPs in detail. (10 Marks)

Module-3

- 5 a. Explain the transport and tunnel modes in IP security. (10 Marks)
b. Mention the applications and benefits of IP security. (10 Marks)

OR

- 6 a. Illustrate the working of basic combinations of security associations. (10 Marks)
b. Explain the Internet key Exchange process using Diffie-Hellman algorithm. (10 Marks)

Module-4

- 7 a. Explain the 3 classes of intruders with examples and explain the intruder behavior patterns. (10 Marks)
b. Explain the types of malicious software in detail. (10 Marks)

OR

- 8 a. Describe the generations of antivirus software. (10 Marks)
b. With neat diagram, explain the digital immune system approach of virus protection. (10 Marks)

Module-5

- 9 a. What are the capabilities and limitations of firewall? (10 Marks)
b. What are the different types of firewall? With a neat diagram, describe the working of packet filtering firewall. (10 Marks)

OR

- 10 a. With neat diagram, explain the distributed firewall configuration. (08 Marks)
b. Discuss the characteristics of Bastion host. Explain the host based and personal firewalls. (12 Marks)

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BMATEC301/BEC301/BBM301

Third Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 AV Mathematics III for EC/ BM Engineering

Time: 3 hrs.

Max. Marks: 100

- Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. Statistical table and Mathematics formula handbook are allowed.
3. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C																		
Q.1	a.	Obtain the Fourier series of $f(x) = \frac{\pi - x}{2}$ in $0 < x < 2\pi$. Hence deduce that $1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots = \frac{\pi}{4}$.	6	L2	CO1																		
	b.	Find the Fourier series of $f(x) = x $ in $(-\ell, \ell)$. Hence show that $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots = \frac{\pi^2}{8}$.	7	L3	CO1																		
	c.	Expand $f(x) = 2x - 1$ as a cosine half range Fourier series in $0 < x < 1$.	7	L2	CO1																		
OR																							
Q.2	a.	Find the Fourier series of $f(x) = \begin{cases} 1 + \frac{2x}{\pi} & \text{in } -\pi < x < 0 \\ 1 - \frac{2x}{\pi} & \text{in } 0 < x < \pi \end{cases}$. Hence deduce that $\frac{\pi^2}{8} = \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots$.	6	L2	CO1																		
	b.	Obtain the sine half range series of, $f(x) = \begin{cases} \frac{1}{4} - x & \text{in } 0 < x < \frac{1}{2} \\ x - \frac{3}{4} & \text{in } \frac{1}{2} < x < 1 \end{cases}$	7	L2	CO1																		
	c.	Determine the constant term and the first cosine and sine terms of the Fourier series expansion of y from the following data : <table><tr><td>x°:</td><td>0</td><td>45</td><td>90</td><td>135</td><td>180</td><td>225</td><td>270</td><td>315</td></tr><tr><td>y:</td><td>2</td><td>$\frac{3}{2}$</td><td>1</td><td>$\frac{1}{2}$</td><td>0</td><td>$\frac{1}{2}$</td><td>1</td><td>$\frac{3}{2}$</td></tr></table>	x°:	0	45	90	135	180	225	270	315	y:	2	$\frac{3}{2}$	1	$\frac{1}{2}$	0	$\frac{1}{2}$	1	$\frac{3}{2}$	7	L1	CO1
x°:	0	45	90	135	180	225	270	315															
y:	2	$\frac{3}{2}$	1	$\frac{1}{2}$	0	$\frac{1}{2}$	1	$\frac{3}{2}$															
Module – 2																							
Q.3	a.	Find the Fourier transform of the function, $f(x) = \begin{cases} 1 & \text{for } x \leq a \\ 0 & \text{for } x > a \end{cases}$. Hence evaluate $\int_0^\infty \frac{\sin x}{x} dx$.	6	L2	CO2																		
	b.	Find the Fourier sine and cosine transforms of $f(x) = e^{-\alpha x}$, $\alpha > 0$.	7	L2	CO2																		
	c.	Find the Fourier sine transform of $\frac{e^{-ax}}{x}$, $a > 0$.	7	L3	CO2																		

OR																		
Q.4	a.	If $f(x) = \begin{cases} 1-x^2, & x < 1 \\ 0, & x \geq 1 \end{cases}$, find the Fourier transform of $f(x)$ and hence find the value of, $\int_0^{\infty} \frac{x \cos x - \sin x}{x^3} dx$.	6	L2	CO2													
	b.	Find the Fourier sine transform of $f(x) = e^{- x }$ and hence evaluate $\int_0^{\infty} \frac{x \sin mx}{1+x^2} dx$, $m > 0$.	7	L3	CO2													
	c.	Find the Discrete fast fourier of signal $= (0, 1, 49)^T$	7	L3	CO2													
Module – 3																		
Q.5	a.	Find the z-transform of, (i) $\cosh n\theta$ (ii) $\sinh n\theta$	6	L1	CO3													
	b.	If $V(z) = \frac{2z^2 + 3z + 12}{(z-1)^4}$, evaluate u_0, u_1 and u_2	7	L2	CO3													
	c.	Find the inverse z-transform of, $\frac{z}{(z-1)(z-2)}$.	7	L2	CO3													
OR																		
Q.6	a.	Solve by using z-transforms, $y_{n+2} + 2y_{n+1} + y_n = n$ with $y_0 = 0 = y_1$	6	L3	CO3													
	b.	Find $z^{-1} \left[\frac{5z}{(3z-1)(2-z)} \right]$.	7	L2	CO3													
	c.	Solve by using z-transforms $u_{n+2} - 5u_{n+1} + 6u_n = 2^n$ with $u_0 = 0 = u_1$.	7	L3	CO3													
Module – 4																		
Q.7	a.	Solve $\frac{d^3 y}{dx^3} + 6\frac{d^2 y}{dx^2} + 11\frac{dy}{dx} + 6y = 0$.	6	L1	CO4													
	b.	Solve $(D^2 + 1)y = x^2 + 4x - 6$.	7	L2	CO4													
	c.	Using the method of variation of Parameters of $\frac{d^2 y}{dx^2} - 6\frac{dy}{dx} + 9y = e^{3x}$	7	L3	CO4													
OR																		
Q.8	a.	Solve $6\frac{d^2 y}{dx^2} + 17\frac{dy}{dx} + 12y = e^{-x}$.	6	L2	CO4													
	b.	Solve the Cauchy's differential equation, $x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + 8y = 65 \cos(\log x)$.	7	L2	CO4													
	c.	The charge q in a series circuit containing an Inductance L , Capacitance C , emf E satisfy the differential equation, $L \frac{d^2 q}{dt^2} + \frac{q}{C} = E$. Express q in terms of t .	7	L3	CO4													
Module – 5																		
Q.9	a.	Fit a second degree parabola $y = a + bx + cx^2$ into least square sense for the data and estimate y at $x = 6$. <table><tr><td>x:</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>y:</td><td>10</td><td>12</td><td>13</td><td>16</td><td>19</td></tr></table>	x:	1	2	3	4	5	y:	10	12	13	16	19	6	L1	CO5	
x:	1	2	3	4	5													
y:	10	12	13	16	19													

	b.	Find a correlation coefficient for the two variables x and y. <table><tr><td>x:</td><td>92</td><td>89</td><td>87</td><td>86</td><td>83</td><td>77</td><td>71</td><td>63</td><td>53</td><td>50</td></tr><tr><td>y:</td><td>86</td><td>83</td><td>91</td><td>77</td><td>68</td><td>85</td><td>52</td><td>82</td><td>37</td><td>57</td></tr></table>	x:	92	89	87	86	83	77	71	63	53	50	y:	86	83	91	77	68	85	52	82	37	57	7	L2	CO5
x:	92	89	87	86	83	77	71	63	53	50																	
y:	86	83	91	77	68	85	52	82	37	57																	
	c.	Ten students got the following percentage of marks in two subjects x and y. Compute the rank correlation coefficient. <table><tr><td>x:</td><td>78</td><td>36</td><td>98</td><td>25</td><td>75</td><td>82</td><td>90</td><td>62</td><td>65</td><td>39</td></tr><tr><td>y:</td><td>84</td><td>51</td><td>91</td><td>60</td><td>68</td><td>62</td><td>86</td><td>58</td><td>53</td><td>47</td></tr></table>	x:	78	36	98	25	75	82	90	62	65	39	y:	84	51	91	60	68	62	86	58	53	47	7	L2	CO5
x:	78	36	98	25	75	82	90	62	65	39																	
y:	84	51	91	60	68	62	86	58	53	47																	
OR																											
Q.10	a.	If θ is the angle between the lines of regression show that $\tan \theta = \frac{\sigma_x \sigma_y}{\sigma_x^2 + \sigma_y^2} \left(\frac{1 - r^2}{r} \right).$	6	L2	CO5																						
	b.	Obtain the lines of regression and hence find the coefficient of correlation for the data, <table><tr><td>x:</td><td>1</td><td>3</td><td>4</td><td>2</td><td>5</td><td>8</td><td>9</td><td>10</td><td>13</td><td>15</td></tr><tr><td>y:</td><td>8</td><td>6</td><td>10</td><td>8</td><td>12</td><td>16</td><td>16</td><td>10</td><td>32</td><td>32</td></tr></table>	x:	1	3	4	2	5	8	9	10	13	15	y:	8	6	10	8	12	16	16	10	32	32	7	L2	CO5
x:	1	3	4	2	5	8	9	10	13	15																	
y:	8	6	10	8	12	16	16	10	32	32																	
	c.	If $8x - 10y + 66 = 0$ and $40x - 18y = 214$ are the two regression lines. Find $\bar{x}$, $\bar{y}$ and r. Find σ_y if $\sigma_x = 3$.	7	L2	CO5																						

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Module – 3

Q.5	a.	Develop the characteristic equation for i) SR flip flop ii) JK flip flop iii) D flip flop iv) T flip flop.	10	L3	CO3
	b.	Explain serial in, parallel at unidirectional shift register and parallel in serious out unidirectional shift register.	10	L2	CO3

OR

Q.6	a.	Explain Mod-4 ring counter and Mod-8 twisted ring counter with logic diagram and counting sequence.	10	L2	CO3
	b.	Design a synchronous Mod-6 counter using clocked D-flip flop.	10	L3	CO3

Module – 4

Q.7	a.	Explain logical operators and relational operators used in verilog.	8	L2	CO4
	b.	Illustrate i) NETS ii) Register iii) Vector iv) integer data types with an example.	8	L2	CO4
	c.	Write a verilog code for full adder using data flow description style.	4	L2	CO4

OR

Q.8	a.	Illustrate the structure of behavioural description with an example using half adder.	8	L2	CO4
	b.	Illustrate the structure of verilog module with an example using half subtractor.	8	L2	CO4
	c.	Write a verilog code for binary to gray using behavioural description style.	4	L2	CO4

Module – 5

Q.9	a.	Write the syntax of IF and EISE-IF with an example.	8	L2	CO4
	b.	Write logic symbol, flowchart and program for D-latch using behavioural description style.	8	L2	CO4
	c.	Write a verilog code for 8:1 MUX using behavioural description style.	4	L2	CO4

OR

Q.10	a.	Explain the structure of structural model with built in gates using example of half adder. Also mention an primitive built in gates.	8	L2	CO4
	b.	Write a verilog code of a 3-bit ripple carry adder using structural description model.	8	L2	CO4
	c.	Write a verilog code of SR flip flop using behavioural description style.	4	L2	CO4

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Third Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025
Electronic Principles and Circuits

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

Module – 1			M	L	C
Q.1	a.	Derive expressions V_{in} , V_{out} and A_V for a common emitter circuit with ac equivalent circuit with π – model.	12	L1	CO2
	b.	What is the voltage gain and output voltage across the load resistor of V_{DB} amplifier? $R_1 = 10\text{ k}\Omega$, $R_2 = 2.2\text{ k}\Omega$, $R_C = 3.6\text{ k}\Omega$, $R_E = 1\text{ k}\Omega$, $R_L = 10\text{ k}\Omega$, $V_{CC} = 10\text{ V}$, $V_{BE} = 0.7\text{ V}$ and $V_{in} = 2\text{ mV}$.	8	L1	CO1
OR					
Q.2	a.	With a neat diagram, explain loading effect of input impedance.	10	L1	CO1
	b.	Explain three types of Bias circuit, (i) Emitter feedback bias. (ii) Collector feedback bias and (iii) Collector and emitter feedback.	10	L1	CO1
Module – 2					
Q.3	a.	Explain the three biasing methods to bias MOS amplifiers with neat circuit diagram.	10	L2	CO2
	b.	Explain the T-equivalent circuit model of MOSFET.	10	L3	CO2
OR					
Q.4	a.	With a small signal equivalent model of MOSFET, derive an expression of voltage gain and transconductance.	10	L2	CO2
	b.	Explain common source follower and derive the expression of voltage gain with necessary equation.	10	L2	CO2
Module – 3					
Q.5	a.	Explain R and 2R resistor Digital to Analog converter and also derive the expression of output voltage.	10	L2	CO3
	b.	With a neat circuit diagram, explain the operation of Monostable multivibrator.	10	L2	CO3
OR					
Q.6	a.	With a neat diagram, explain operation of RC-phase shift oscillator using op-amp. Write the expression for frequency of oscillations.	8	L2	CO3
	b.	With a net diagram, explain operation of crystal oscillator using BJT and Write necessary equations.	6	L2	CO3
	c.	A crystal has these values $L = 3\text{ H}$, $C_s = 0.05\text{ PF}$, $R = 2\text{ k}\Omega$ and $C_m = 10\text{ PF}$. What are the series and parallel resonant frequencies of the crystal?	6	L3	CO3

Module – 4					
Q.7	a.	Explain the first order Low Pass filter with frequency response.	10	L2	CO4
	b.	Explain the two types of Band Pass filters.	10	L2	CO4
OR					
Q.8	a.	Explain the four types of Negative feedback circuits.	10	L2	CO4
	b.	Explain the working of 2 nd order high pass filter with a neat circuit and frequency response.	10	L2	CO4
Module – 5					
Q.9	a.	Explain two load lines with necessary circuit diagram and equations.	10	L2	CO5
	b.	With a neat diagram, explain the working of a Thyristor.	10	L2	CO5
OR					
Q.10	a.	Explain Basic Construction and working of IGBTs with necessary figure.	10	L2	CO5
	b.	With a neat diagram, explain the working of UJT relaxation oscillator.	10	L2	CO5

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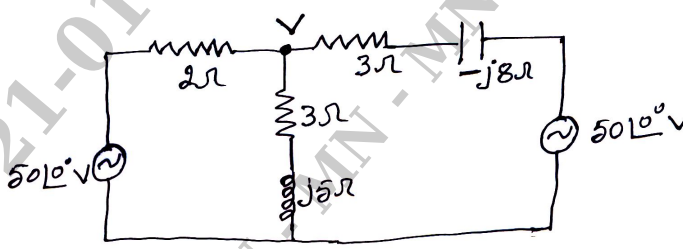
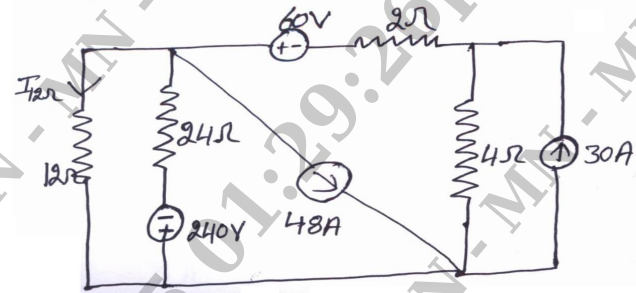
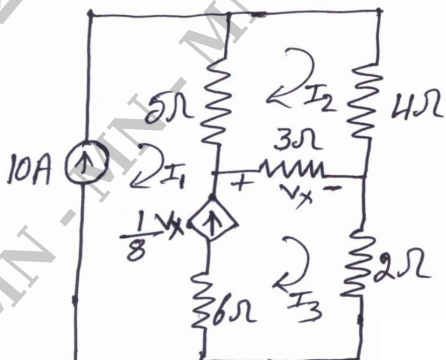
Third Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

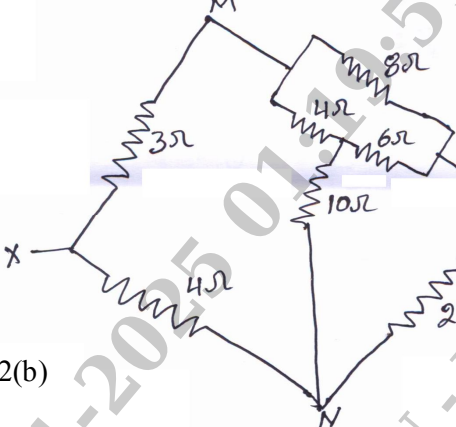
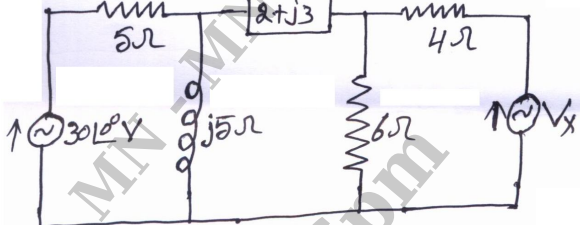
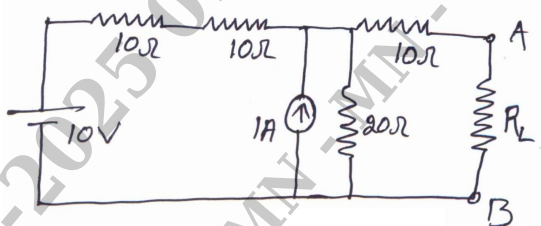
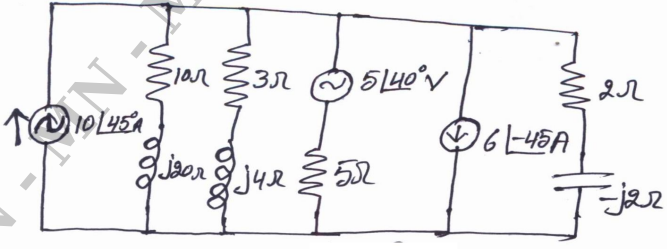
Network Analysis

Time: 3 hrs.

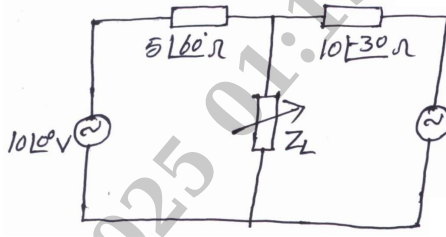
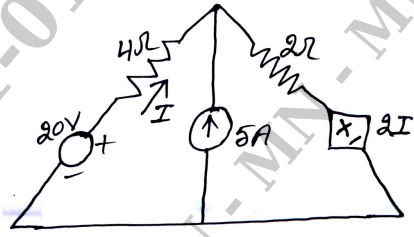
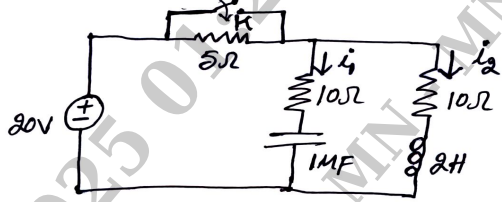
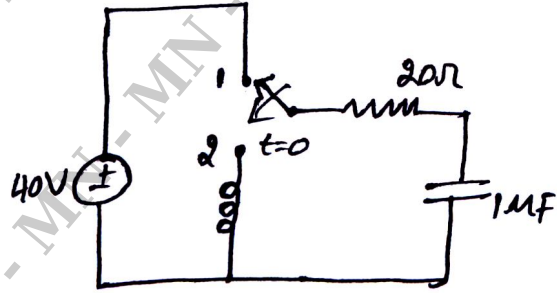
Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

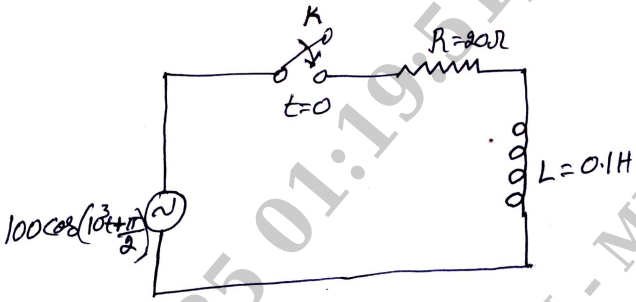
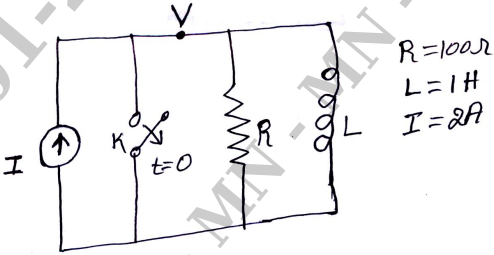
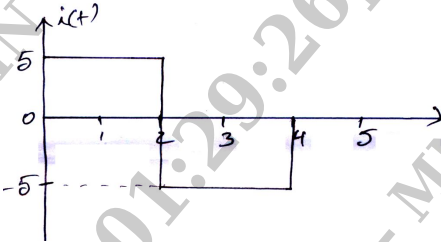
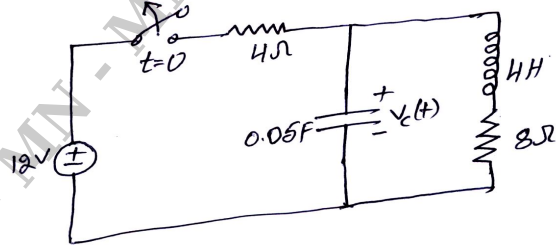
Module – 1			M	L	C
Q.1	a.	Three impedances are connected in Delta. Obtain the star equivalent of the network.	7	L3	CO1
	b.	For the circuit shown in Fig. Q1(b). Find the voltage 'V' at node by using nodal analysis. Fig. Q1(b) 	6	L3	CO1
	c.	Determine the current in 12Ω resistor shown in Fig. Q1(c) using source transformation method. Fig. Q1(c) 	7	L3	CO1
OR					
Q.2	a.	Find the loop currents I_1 , I_2 , and I_3 in the circuit shown in Fig. Q2(a). Fig. Q2(a) 	7	L3	CO1

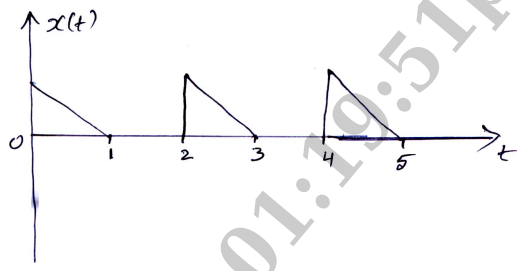
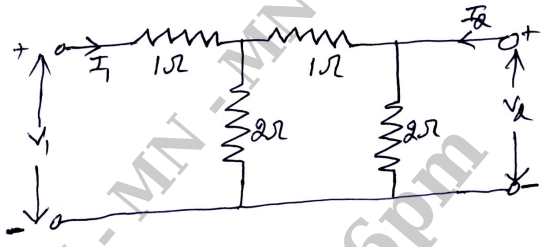
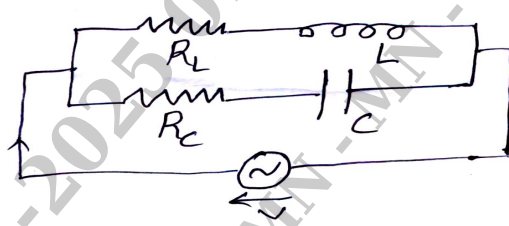
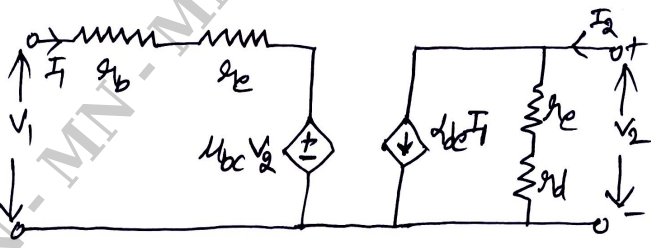
	b. Determine the resistance between the terminals X, Y using star delta transformation in the network shown in Fig. Q2(b).  Fig. Q2(b)	6	L3	CO1
	c. Use the nodal analysis to find the value of V_X and the circuit shown in Fig. Q2(c). Such that the current through $(2 + j3) \Omega$ Impedance is zero.  Fig. Q2(c)	7	L3	CO1
Module – 2				
Q.3	a. State and prove Superposition theorem.	7	L2	CO2
	b. For the circuit shown in Fig. Q3(b), obtain the Thevenin's equivalent circuit.  Fig. Q3(b)	7	L3	CO2
	c. Using Millman's theorem, find current flowing through $(3 + j4) \Omega$ impedance for the circuit shown in Fig. Q3(c).  Fig. Q3(c)	6	L3	CO2

OR

Q.4	a.	State and prove Norton's theorem.	7	L2	CO2
	b.	Find the value of Z_L for Maximum Power transfer and the value of Maximum power for the circuit shown in Fig. Q4(b).	6	L3	CO2
		Fig. Q4(b) 			
	c.	Find current 'I' using Super position theorem for the circuit shown in Fig. Q4(c).	7	L3	CO2
		Fig. Q4(c) 			
Module – 3					
Q.5	a.	Use the concepts of initial condition to illustrate the voltage behavior in inductor circuit for DC supply.	6	L3	CO3
	b.	In the circuit steady state is reached with switch 'K' open. The switch is closed at $t = 0$. Compute i , di/dt and d^2i/dt^2 at $t = 0^+$.	7	L3	CO3
		Fig. Q5(b) 			
	c.	The switch is moved from position (1) to position (2) at $t = 0$. The steady state has been reached before switching. Computer i , di/dt and d^2i/dt^2 at $t = 0^+$ for Fig. Q5(c).	7	L4	CO3
		Fig. Q5(c) 			

OR

Q.6	a. In the circuit shown in Fig. Q6(a), determine complete solution for current when switch 'K' is closed at $t = 0$. Fig. Q6(a) 	10	L3	CO3
	b. Compute v, dv/dt, d^2v/dt^2 at $t = 0^+$ for the circuit shown in below Fig. Q6(b), when the switch K is opened at $t = 0$. Fig. Q6(b) 	10	L4	CO3
Module – 4				
Q.7	a. Using waveform synthesis method to express the voltage pulse in terms of unit step. Find i) $L\{i(t)\}$ ii) $L\{\int i(t).dt\}$. Fig. Q7(a) 	8	L3	CO4
	b. State and prove initial value and final value theorem for Laplace transform.	6	L2	CO4
	c. Obtain the Laplace transform of step and ramp function with relevant expressions.	6	L3	CO4
OR				
Q.8	a. Determine $i_L(t)$ for $t \geq 0$ using Laplace transform for circuit shown in Fig. Q8(a). Fig. Q8(a) 	10	L3	CO4

	b.	Find the Laplace transform of the periodic signal $x(t)$ as shown in Fig. Q8(b).	10	L3	CO4
		 Fig. Q8(b)			
Module – 5					
Q.9	a.	Define Z – parameters. Determine Y parameters in terms of Z – parameters.	6	L3	CO5
	b.	Show that resonant frequency is geometric mean of cut off frequency in series R – L – C circuit.	7	L3	CO5
	c.	Apply the two – port network analysis technique to determine ABCD – parameters of the network shown in Fig. Q9(c).	7	L3	CO5
		 Fig. Q9(c)			
OR					
Q.10	a.	Derive the expression for the resonant frequency of the circuit shown in Fig. Q10(a). Also show that the circuit resonates at all frequency if $R_L = R_C = \sqrt{\frac{L}{C}}$.	10	L3	CO5
		 Fig. Q10(a)			
	b.	The model of a transistor in the CE mode is shown in Fig. Q10(b). Determine the h – parameters.	10	L3	CO5
		 Fig. Q10(b)			

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Third Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

Computer Organization and Architecture

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.

2. M : Marks , L: Bloom's level , C: Course outcomes.

Module – 1			M	L	C
Q.1	a.	With neat diagram explain connection between the processor and memory.	10	L1	CO1
	b.	Write the difference between little endian and big endian memory assignments.	05	L1	CO1
	c.	Write a short note on basic performance equation.	05	L1	CO1
OR					
Q.2	a.	Describe the concept of branching with an example program of instruction execution.	10	L1	CO1
	b.	Represent the following decimal values as signed 7-bit numbers using sign and magnitude, signed 1's complement and signed 2's complement formats. – 55, +51, 8, – 27, – 39, +43, – 10, 62	05	L2	CO1
	c.	Write a short note on memory operations.	05	L1	CO1
Module – 2					
Q.3	a.	What is an addressing mode? Explain any four types of addressing modes, with suitable example.	10	L1	CO2
	b.	Write a program to compute the sum of test scores of all the students in the three tests. Store the corresponding sums in memory.	10	L2	CO2
OR					
Q.4	a.	Explain the Rotate and Shift instructions with an example.	10	L1	CO2
	b.	Define subroutine. Explain subroutine linkage using a link register.	05	L1	CO2
	c.	What are assembler directives? Explain any two directives.	05	L1	CO2
Module – 3					
Q.5	a.	Define I/O interface? Explain I/O interface to connect an input device to the bus with neat diagram.	10	L1	CO3
	b.	What is interrupt? Discuss interrupt I/O method for data transfer.	05	L1	CO3
	c.	Describe two methods of handling multiple devices.	05	L1	CO3
OR					
Q.6	a.	Explain the use of DMA controllers in a computer system with neat diagram.	10	L1	CO3
	b.	Write a note on Bus Arbitration.	10	L1	CO3
Module – 4					
Q.7	a.	Explain the organization of 1K×1 memory chip.	10	L1	CO4
	b.	Write a note on : (i) Static memories (ii) Cache memory	10	L1	CO4
OR					
Q.8	a.	Explain the Magnetic disk principles.	10	L1	CO4
	b.	Draw and explain the internal organization of 2M×8 asynchronous DRAM chip.	10	L2	CO4
Module – 5					
Q.9	a.	Discuss with neat diagram the single bus organization of data path inside a processor.	10	L1	CO5
	b.	What are the actions required to execute a complete instruction ADD (R ₂), R ₁	10	L1	CO5
OR					
Q.10	a.	Draw and explain multiple bus organization of CPU.	10	L1	CO5
	b.	Draw and explain organization of the control unit to allow conditional branching in the microprogram.	10	L1	CO5

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Fourth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025
Electromagnetic Theory

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

Module – 1			M	L	C
Q.1	a.	State and explain Coulomb's law of force between two point charges in vector form.	8	L1	CO1
	b.	Define Electric field intensity. Derive the expression for the electric field intensity at a point due to infinite line charges (Uniformly charged wire).	8	L2	CO1
	c.	Two very small conducting spheres, each of mass 1×10^{-4} kg are suspended at common point by very thin filaments of length 0.2m. A charge Q Coulomb is placed on each sphere. The electric force of repulsion separates the spheres and an equilibrium is reached when the suspending filaments make an angle of 10° . Assuming $e_r = 1$, $g = 9.8\text{N/kg}$ and negligible mass for the filaments, find Q.	4	L3	CO1
OR					
Q.2	a.	Define Point charge and using Coulomb's Law, derive expression for electric field intensity due to a point charge.	8	L2	CO1
	b.	Let a point $Q_1 = 25\text{nc}$ be located at A(4, -2, 7) and a charge $Q_2 = 60\text{nc}$ be at B(-3, 4, -2). Find $\vec{E}$ at C(1, 2, 3). Also find direction of the electric field. Given $\epsilon_0 = 8.854 \times 10^{-12} \text{ F/m}$.	8	L3	CO1
	c.	Two point charges of $+3 \times 10^{-9} \text{ C}$ and $-2 \times 10^{-9} \text{ C}$ are spaced two meter apart. Determine the electric field at a point which is one meter from each of the two point charges.	4	L3	CO1
Module – 2					
Q.3	a.	State and prove Gauss Divergence theorem or divergence theorem.	8	L2	CO2
	b.	A point charge, $Q = 30\text{nc}$ is located at the origin in Cartesian coordinates. Find the electric flux density and electric field intensity at (1, 3, -4)m.	8	L3	CO2
	c.	Derive an equation for equation of continuity (continuity of current).	4	L3	CO2
OR					
Q.4	a.	State and prove Gauss law.	8	L2	CO2
	b.	Given that the potential field is $V = 2x^2y - 5z$. Find the potential, electric field intensity and volume charge density at point P(-4, 3, 6).	8	L3	CO2
	c.	State Gauss law in point form. Hence derive Maxwell's first equation.	4	L3	CO2

Module – 3					
Q.5	a.	Starting from gauss law, derive Poisson's and Laplace equation. Hence define Laplace equation in all three coordinate systems.	4	L2	CO3
	b.	State and prove Stoke's theorem.	8	L2	CO3
	c.	Find the potential and volume charge density at P(0.5 , 1.5 , 1)m in free space. Given the potential field as under. i) $V = 2x^2 - y^2 - z^2$ volt ii) $V = 6 r \phi z$ volt.	8	L3	CO3
OR					
Q.6	a.	State and prove Biot – Savart's law.	4	L1	CO3
	b.	State and prove Ampere's circuital law.	8	L1	CO3
	c.	The magnetic field intensity is given in a certain region of space as : $\vec{H} = \left(\frac{x+2y}{z^2} \right) \hat{a}_y + \frac{2}{z} \hat{a}_z$ A/m. i) Find $\nabla \times \vec{H}$ ii) Find $\vec{J}$ iii) Use $\vec{J}$ to find total current passing through the surface , $Z = 4$, $1 < x < 2$, $3 < y < 5$ in the $\hat{a}_z$ direction.	8	L3	CO3
Module – 4					
Q.7	a.	Define current element. Derive an equation for force on a differential current element in a magnetic field.	8	L2	CO4
	b.	A point charge $Q = 18\text{nc}$ has a velocity of 5×10^6 m/s in the direction $\vec{a} = 0.6 \hat{a}_x + 0.75 \hat{a}_y + 0.3 \hat{a}_z$. Calculate the magnitude of the force exerted on the charge by the field $\vec{B} = -3 \hat{a}_x + 4 \hat{a}_y + 6 \hat{a}_z$ mT.	8	L3	CO4
	c.	Calculate the force on a straight conductor of length 0.3m carrying a current 5A in the Z – direction where the magnetic field is $\vec{B} = 3.5 \times 10^{-3} (a\hat{x} - a\hat{y})$ Tesla. ($a\hat{x}$ and $a\hat{y}$ are unit vectors).	4	L3	CO4
OR					
Q.8	a.	Derive magnetic boundary condition for i) Tangential component of magnetic field. ii) Normal component of magnetic field.	8	L2	CO4
	b.	A conductor 4m long lies along the Y – axis with a current of 10A in the $a\hat{y}$ direction. Find the force on the conductor if the field in the region is $\vec{B} = 0.05 a\hat{x}$ tesla.	8	L3	CO4
	c.	Find the magnetic field intensity inside a magnetic material for following conditions : $M = 100\text{A/m}$ and $\mu = 1.5 \times 10^{-5}$ H/m $B = 200\mu\text{T}$, X_m (Magnetic susceptibility = 15).	4	L3	CO4

Module – 5					
Q.9	a.	Derive Integral and point form of Faraday's law.	8	L2	CO5
	b.	Given $\vec{E} = E_m \sin (wt - \beta z) \hat{a}_y$ in free space. Calculate $\vec{D}$, $\vec{B}$ and $\vec{H}$.	8	L3	CO5
	c.	A copper disc 40cm diameter is rotated at 3000 r.p.m on a horizontal axis perpendicular to and through the centre of disc axis, lying in magnetic meridian. Two brushes make contact with the disc at diametrically opposite points on the edge. If horizontal component of earth's field is 0.02 mT, find the induced e.m.f between brushes.	4	L3	CO5
OR					
Q.10	a.	State and derive Poynting's theorem for uniform plane waves.	8	L2	CO5
	b.	Derive general wave equation in electric and magnetic fields.	8	L2	CO5
	c.	For silver, the conductivity is $\sigma = 3.0 \times 10^6$ s/m. At what frequency will depth of penetration be 1mm?	4	L3	CO5

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Fourth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

Principles of Communication Systems

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	Define probability. Illustrate the relationship between sample space, events and probability.	06	L1	CO1
	b.	Outline random processes and illustrate an ensemble of sample function with a neat diagram.	06	L2	CO1
	c.	Show that if a Gaussian process $x(t)$ is applied to a stable linear filter, then the random process $y(t)$ developed at the output of the filter is also Gaussian.	08	L3	CO2
OR					
Q.2	a.	What is conditional probability? Prove that $P(B/A) = P(A/B) \cdot P(B) / P(A)$	06	L1	CO1
	b.	Define mean, correlation and covariance function.	06	L2	CO2
	c.	Develop a program to generate the probability density function of Gaussian distribution function.	08	L3	CO2
Module – 2					
Q.3	a.	An antenna has an impedance of 40Ω an unmodulated AM signal produces a current of 4.8 A. The modulation is 90 percent calculate i) The carrier power ii) The total power iii) The sideband power	06	L1	CO1
	b.	Explain with neat diagrams amplitude demodulation using the diode detector.	07	L1	CO1
	c.	Explain a general block diagram of an FDM system	07	L2	CO2
OR					
Q.4	a.	Interpret the concept of modulation index and percentage of modulation write the necessary equations.	06	L1	CO1
	b.	Explain high level collector modulation with neat block diagram.	07	L2	CO1
	c.	Explain with diagrams the working principle of lattice type balanced modulator.	07	L2	CO2
Module – 3					
Q.5	a.	Compare and contrast FM and AM.	06	L1	CO1
	b.	Explain with diagrams the working principle of frequency modulation using voltage controlled oscillator.	07	L2	CO2
	c.	Explain general block diagram of a super heterodyne receiver.	07	L2	CO2
OR					
Q.6	a.	The input to an FM receiver having an S/N of 2.8. The modulating frequency is 1.5 KHz. The maximum permitted deviation is 4 KHz. What are (i) The frequency deviation caused by the noise (ii) The improved output S/N.	06	L2	CO2
	b.	Define PLL. Explain the basic block diagram of a PLL.	07	L1	CO2
	c.	Explain JFET mixer.	07	L2	CO2

Module – 4					
Q.7	a.	What are the advantages of digital signal over analog signals?	04	L1	CO1
	b.	Explain with basic elements of a PCM system with neat diagrams.	08	L2	CO1
	c.	For the data stream 0 1 1 0 1 0 0 1 draw the following line code waveforms i) Unipolar NRZ ii) Polar NRZ iii) Unipolar RZ iv) Manchester code	08	L3	CO2
OR					
Q.8	a.	State and prove Sampling theorem.	04	L1	CO1
	b.	What is multiplexing and why is it required in communication? Explain the working of TDM with a neat block diagram.	08	L2	CO1
	c.	Explain the generation of PPM with a relevant block diagrams and waveforms.	08	L2	CO2
Module – 5					
Q.9	a.	Define Intersymbol interference (ISI) outline baseband binary data transmission system with neat block diagram and equations.	08	L2	CO1
	b.	Develop a code to generate RZ pulse.	04	L3	CO2
	c.	Define signal to noise ratio. Explain different types of external and internal noise.	08	L2	CO1
OR					
Q.10	a.	Explain the following concept briefly: i) Nyquist criterion for distributors transmission ii) Baseband M-ary PAM transmission	08	L1	CO2
	b.	Develop a code to generate Raised cosine pulse.	04	L2	CO2
	c.	Illustrate the concept of noise in cascaded stages with a diagram. Write Friis formula and mention its terms.	08	L2	CO3

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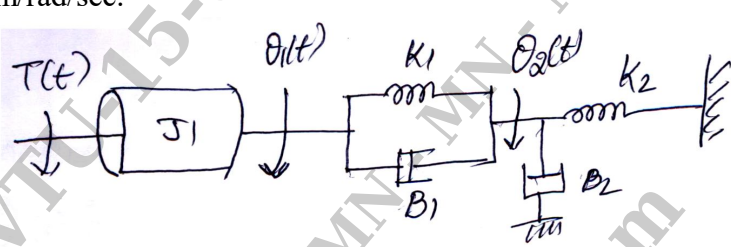
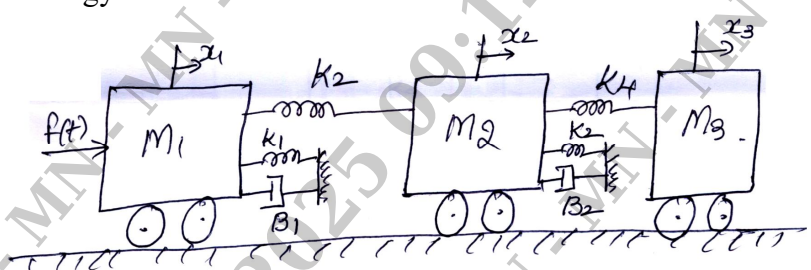
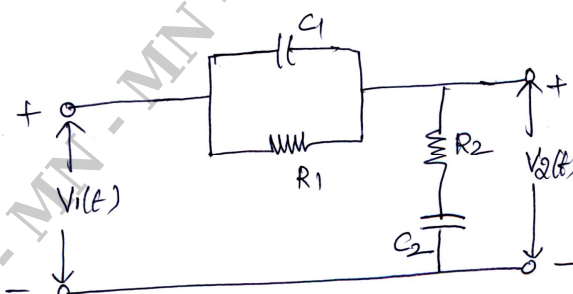
Fourth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

Control Systems

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	Compare open loop and closed loop control system with practical example.	06	L2	CO1
	b.	For the system shown in Fig.Q1(b). Find the transfer function $G(s) = \frac{\theta_2(s)}{T(s)}$ consider $J_1 = 1 \text{ kgm}^2$, $K_1 = 1 \text{ Nm/rad}$, $K_2 = 1 \text{ Nm/rad}$, $B_1 = 1 \text{ Nm/rad/sec}$, $B_2 = 1 \text{ Nm/rad/sec}$.  Fig.Q1(b)	06	L2	CO1
	c.	Draw the mechanical network for the system shown in Fig.Q1(c). Write the equations of performance and draw its analogous circuit based one force voltage analogy.  Fig.Q1(c)	08	L2	CO1
OR					
Q.2	a.	The circuit shown in Fig.Q2(a) is called lead-lag filter. Find the transfer function $\frac{V_2(s)}{V_1(s)}$ when $R_1 = 100 \Omega$, $R_2 = 200 \text{ K}\Omega$, $C_1 = 1 \mu\text{F}$ and $C_2 = 0.1 \mu\text{F}$.  Fig.Q2(a)	10	L3	CO1

- b. What are the variables and elements of translational motion? For the mechanical system shown in Fig.Q2(b).
- Write the differential equations of performance.
 - Draw and write loop and nodal equations based on F-V and F-I analogous networks.

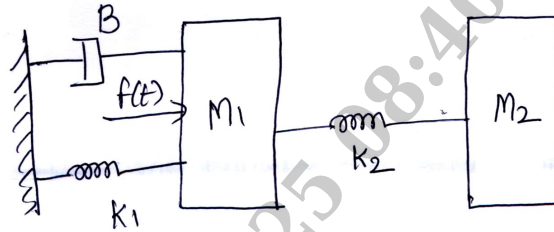


Fig.Q2(b)

Module – 2

- Q.3 a. Give any six block diagram reduction rules to find the transfer function of the system. **04** **L1** **CO2**

- b. For the system represented in the given Fig.Q3(b), determine transfer function $C(s)/R(s)$. **06** **L2** **CO1**

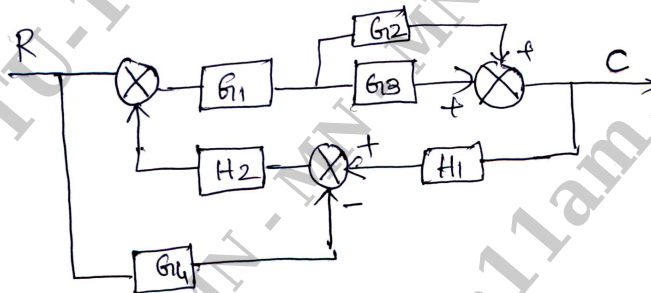


Fig.Q3(b)

- c. Find the overall transfer function of the system whose signal flow graph is shown in Fig.Q3(c). **10** **L2** **CO2**

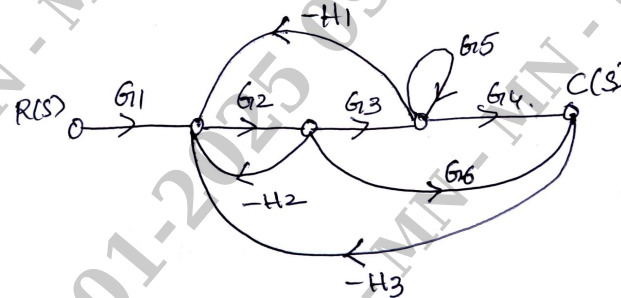


Fig.Q3(c)

OR

- Q.4 a. Interpret the transfer function by converting the block diagram into signal flow graph. **10** **L2** **CO2**

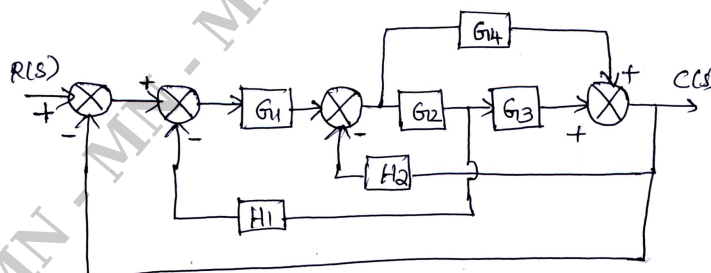


Fig.Q4(a)

- b.** Obtain the transfer function for the block diagram shown in Fig.Q4(b) using block diagram reduction technique.

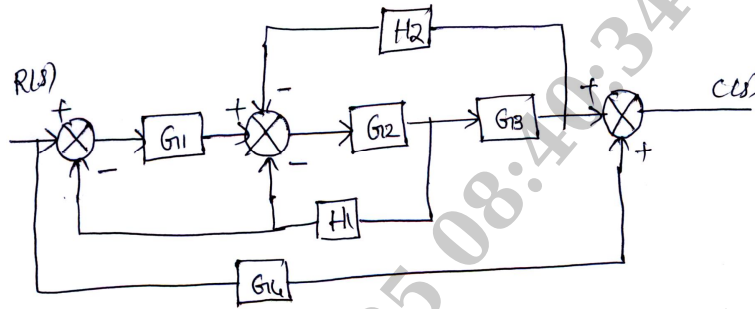


Fig.Q4(b)

Module – 3

- Q.5 a.** Make use of the response curve of 2nd order under-damped system to define and derive the expression for (i) peak time (ii) peak overshoot (iii) rise time **10 L2 CO3**
- b.** Find K_p , K_v and K_a for a system having $G(s) = \frac{s+10}{s(s^3+7s^2+12s)}$. Also, evaluate the steady state error, when the I/P $r(t)$ is given by:
 (i) $r(t) = 5u(t)$ (ii) $r(t) = 2t u(t)$ (iii) $r(t) = 4t^2 u(t)$ **10 L2 CO3**

OR

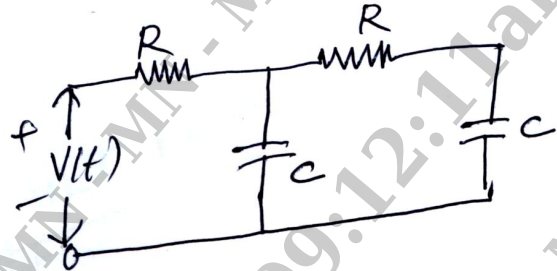
- Q.6 a.** Derive an expression for the under damped response of a second order feedback control system for step input. **10 L2 CO2**
- b.** Explain the static error constant and derive the expressions. **06 L2 CO2**
- c.** Analyze the effect of PD controller for 2nd order control system with appropriate equations. **04 L2 CO2**

Module – 4

- Q.7 a.** The open loop transfer function of a unity feedback system is given by $G(s) = \frac{K}{s(s+3)(s^2+s+1)}$. Find the value of K that will cause sustained oscillation and hence find the oscillation frequency. **08 L2 CO3**
- b.** Sketch the root locus plot for a negative feedback control system whose open loop transfer function is given by $G(s)H(s) = \frac{K}{s(s+1)(s+2)(s+3)}$. For all values of K ranging from 0 to ∞ . Find the value of K for closed loop stability. **12 L3 CO3**

OR

- Q.8 a.** For the characteristic equations given below, determine number of roots with positive real part: **10 L2 CO4**
 i) $s^6 + s^5 + 3s^4 + 2s^3 + 5s^2 + 3s + 1 = 0$
 ii) $s^8 + s^7 + 4s^6 + 3s^5 + 14s^4 + 11s^3 + 20s^2 + 9s + 9 = 0$

	b.	Show that the part of root locus of a system with $G(s)H(s) = \frac{K(s+3)}{s(s+2)}$ is a circle having center $(-3, 0)$ and radius at $\sqrt{3}$.	10	L3	CO3
Module – 5					
Q.9	a.	Construct the bode plot for the transfer function $G(s) = \frac{80}{s(s+2)(s+20)}$. Determine GM and PM, ω_{pc} , ω_{gc} .	10	L2	CO3
	b.	Obtain the state transition matrix for the following system: $\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -1 & -0.5 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0.5 \\ 0 \end{bmatrix} u$	10	L2	CO5
OR					
Q.10	a.	Using Nyquist stability criteria investigate the stability negative feedback control system whose open loop transfer function is given by $G(s)H(s) = \frac{100}{(s+1)(s+2)(s+3)}$. Assume $\omega_g = 1.253$ rad/sec.	10	L2	CO5
	b.	Obtain the state model of electrical network shown in Fig.Q10(b), by choosing $V_1(t)$ and $V_2(t)$ as state variables.  Fig.Q10(b)	10	L3	CO5

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Fourth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

Microcontrollers

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	Bring out the difference between Microprocessor and Microcontroller.	04	L2	CO1
	b.	With function of each pin, explain the pin layout of 8051 Microcontroller.	10	L2	CO1
	c.	Summarize the internal RAM configuration of 8051.	06	L2	CO1
OR					
Q.2	a.	Differentiate between CISC and RISC processor architectures.	04	L2	CO1
	b.	With a neat architecture, explain the architectural features of 8051.	08	L2	CO1
	c.	Interface 8051 microcontroller to 16K bytes of EPROM and 8K bytes of RAM. Explain with neat sketch.	08	L3	CO1
Module – 2					
Q.3	a.	What is an addressing mode? Explain 4 different addressing modes of 8051 with examples.	08	L2	CO2
	b.	Illustrate with a neat diagram different ranges of jump instructions.	06	L2	CO2
	c.	Write an ALP to convert a packed BCD number into two ASCII numbers. Store the result in R5 and R6 respectively.	06	L2	CO2
OR					
Q.4	a.	Define assembler directives. Explain the same with examples.	08	L2	CO2
	b.	List and explain bit level logical instructions in 8051.	06	L2	CO2
	c.	Develop an assembly language program to swap the contents of R3 and R4 registers in BANK0 using different methods.	06	L2	CO2
Module – 3					
Q.5	a.	Explain the bit contents of TCON and TMOD registers.	06	L2	CO3
	b.	Develop an ALP to generate a square wave of frequency 1 kHz on Pin P1.2 using Timer 0 in mode 2. Show the delay calculation. Assume XTAL frequency = 22 MHz.	06	L3	CO3
	c.	Explain RS232 in serial communication using 8051 Microcontroller with DB-9 pin connector.	08	L2	CO3
OR					
Q.6	a.	Explain the bit pattern of SCON register with diagram.	04	L2	CO3
	b.	Develop an 8051 C program to transfer letter "A" serially at 9600 baud rate, 8 bit data, 1 stop bit, do this continuously.	08	L3	CO3
	c.	Explain Mode 2 operations of timers and explain steps involved in programming timer in Mod 2, with necessary diagram.	08	L2	CO3
Module – 4					
Q.7	a.	Explain the structure of interrupt priority and interrupt enable register.	08	L2	CO4
	b.	Explain interrupt vector table of 8051 Microcontroller.	06	L2	CO4
	c.	Explain programming of Timer interrupts.	06	L2	CO4
1 of 2					

OR

Q.8	a.	List the steps involved in executing an interrupt.	04	L2	CO4
	b.	Write an ALP program using interrupts to generate a square wave on port pin P1.2 of 10 kHz using timer 0 in mode 2, XTAL = 22 MHz.	08	L3	CO4
	c.	Explain the steps involved in programming serial communication interrupts.	08	L2	CO4
Module – 5					
Q.9	a.	With a neat diagram, write an 'C' language program to interface DAC to 8051 Microcontroller to generate staircase waveform with 20 steps.	10	L3	CO5
	b.	Explain the interfacing of DC motor using C programming.	10	L3	CO5
OR					
Q.10	a.	With neat diagram, write an C language program to interface stepper motor to 8051 Microcontroller.	10	L3	CO5
	b.	Write a C program to display 'HELLO WORLD' by interfacing LCD display to 8051 Microcontroller.	10	L3	CO5

CBCS SCHEME

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BBOK407

Fourth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

Biology for Engineers

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.

2. M : Marks , L: Bloom's level , C: Course outcomes.

Module – 1			M	L	C
Q.1	a.	Define cell. Explain the structure and function of plant cell with neat diagram.	08	L2	CO1
	b.	Define Stem Cell. Discuss the types and application of stem cells.	06	L2	CO1
	c.	Describe the properties and functions of hormones.	06	L2	CO1
OR					
Q.2	a.	Discuss the properties and functions of nucleic acids in cellular processes.	07	L2	CO1
	b.	Discuss the properties and functions of enzymes.	07	L2	CO1
	c.	Discuss the properties of vitamins and its supplies.	06	L2	CO1
Module – 2					
Q.3	a.	Apply the knowledge of nucleic acid in DNA finger printing in forensic applications.	08	L3	CO1
	b.	Discuss whey protein and plant based protein as protein based food.	06	L2	CO1
	c.	Write a note on PLA as bioplastic.	06	L1	CO1
OR					
Q.4	a.	Apply your knowledge of lipids and outline the process of obtaining biodiesel from lipids.	07	L3	CO1
	b.	Define vaccine. Discuss the mechanism of RNA vaccine for COVID-19.	07	L2	CO1
	c.	Write a note on enzyme based biosensors.	06	L1	CO1
Module – 3					
Q.5	a.	Compare human brain with computer's CPU.	07	L3	CO2
	b.	Explain lungs as a purification system.	07	L2	CO2
	c.	Write a note on dialysis systems of kidney.	06	L1	CO2
OR					
1 of 2					

Q.6	a.	Illustrate the engineering solutions available for Parkinson's disease.	07	L3	CO2
	b.	Explain heart as a pumping system.	07	L2	CO2
	c.	Write a note on optical correction and materials used for lens.	06	L1	CO2
Module – 4					
Q.7	a.	Illustrate the HBOCs and PFCs as human substituents.	07	L3	CO3
	b.	Explain how the structure of shark skin reduces drag and how these properties have been applied to improve swim suit.	07	L2	CO3
	c.	Explain the term GPS and aircrafts technology as bio inspired by bird fly.	06	L2	CO3
OR					
Q.8	a.	Compare the uses of ultrasonography and sonars.	07	L3	CO3
	b.	Discuss the king fisher beak shaped bullet train to the reduction of noise and improve the stability.	06	L2	CO3
	c.	Explain the term superhydrophobic and self-cleaning in lotus leaf effect.	07	L2	CO3
Module – 5					
Q.9	a.	Explain bioimaging and artificial intelligence technique in disease diagnosis.	08	L2	CO4
	b.	Explain the working principles of electrical tongue and electrical nose in food industry.	06	L2	CO4
	c.	Write a note on bioengineering of Muscular dystrophy and osteoporosis.	06	L1	CO4
OR					
Q.10	a.	Explain the process of biomineralization via microbial surface adsorption.	07	L2	CO4
	b.	Describe the concept of DNA origami and its role in bio-computing.	06	L2	CO4
	c.	Write a note on self healing bio-concrete and bio-mineralization processes.	07	L1	CO4

CBCS SCHEME

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Question Paper Version : A

Fourth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

Universal Human Values

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the **fifty** questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

1. What is the state of liking and a holistic and all-encompassing state of the mind that crates inner harmony?
a) Prosperity b) Happiness c) Innateness d) Organized
2. _____ is defined as a group of people working together to create a surplus
a) Alone b) Organisation c) System d) Army
3. The process whereby national economies and business systems are becoming deeply interlinked with each other is called
a) Nationalization b) Glocalization c) Global linking d) Globalization
4. Which of the following actions will likely lead to organizational ethical behavior?
a) Establishing an ethics office
b) Promoting moral courage
c) Developing strong governance processes
d) All of these
5. _____ behavior tends to arise when managers decide to put the attainment of their own personal goals, or the goals of the organization, above the fundamental rights of one or more stakeholder groups
a) Complementary b) Situational c) Unethical d) Confusing
6. Accepted principles of right or wrong governing the conduct of business people are called :
Choose one answer
a) Business b) Conduct c) Business ethics d) Principles
7. What is true about a professional
a) The professional has to have autonomy
b) He cannot be supervised by the client
c) He should not subject to political control
d) All of these

8. The basic rule of professionalism is “not knowingly to do _____”
a) Harm b) Work c) Competition d) Copying
9. The study of _____, as contrasted with legal, obligations is called ethics
a) Behaviors b) Moral c) Business d) All of these
10. Ethics is a conception of _____ conduct
a) Bad/Wrong b) Good/Correct c) Right/True d) Right and Wrong
11. Ethics and _____ go together
a) Profits b) Management c) Worker ship d) All of these
12. _____ driven companies are always successful in the long run
a) Management b) Worker union c) Value d) Profit
13. A business needs to be ethical, in order to
a) Fulfill public expectations for business
b) Prevent harming others
c) Improve business relations and employees productivity
d) All of these
14. Following is (are) types(s) of ethics issues(s)
a) Functional area ethics b) Corporate policy ethics
c) Face to face ethics d) All of these
15. What is called living with assumption for oneself as body and living of human being only on the basis of physical facilities, and not with right understanding and relationship?
a) Human consciousness b) Happiness
c) Right Understanding d) Animal Consciousness
16. Samridhi means
a) Happiness b) Wealth c) Prosperity d) Health
17. Developed nations are the live example of
a) Prosperity b) Wealth c) Happiness d) Health
18. The participation of human beings is seen in two forms
a) Prosperity and work b) Values and Understanding
c) Behavior and Wealth d) Behavior and Work
19. What are the outcomes of realization and understanding?
a) Work b) Values c) Happiness d) Health
20. Corporate culture _____ refers to how widely and deeply employees hold the company’s dominant values and assumptions
a) Breadth b) Depth c) Character d) Strength
21. The observable symbols and signs of a organization’s culture are called :
a) Its cultural design b) Artifacts c) Cultural formatting d) Its cultural dcor
22. _____ is the feeling or responsibility and commitment for ensuring Right Understanding and Right Feeling in the self of my relative
a) Guidance b) Care c) Respect d) Affection
23. The ability to perform a specific task, action, or function successfully is called _____
a) Competence b) Purpose c) Intention d) Program

24. To live in harmony with other human beings is ____
a) Svatva b) Swatantrata c) Swarajya d) None of these
25. Natural acceptance is?
a) Universal b) Definite c) Continuous d) All of these
26. Basic human aspirations are ____
a) Happiness b) Prosperity c) Education d) Both (a) and (b)
27. Which of the following is true?
a) Happiness is definite and it can be understood
b) I need to be unhappy to recognize that I am happy
c) One will be bored of happiness if one is always happy
d) I will grow only if I am unhappy. If I become happy my growth will stop
28. To be wealthy is ____ condition in the modern world to be happy
a) Not essential b) True c) Fake d) The most important
29. It is important to have a good ____ to be successful in your career
a) Appear b) Attend c) Attitude d) Respectful
30. ____ skills such as writing, speaking and body language are important work ethics so that you can be clear in your messages to others
a) Communication b) Cooperation c) Honesty d) Integrity
31. You are a person with a high level of ____ if you stand by you stand by your values even if others put pressure on you to do something that is wrong or if they make fun of you
a) Communication b) Cooperation c) Honesty d) Integrity
32. If you don't have ____ you can really get in trouble for lying, cheating or deceiving others
a) Communication b) Cooperation c) Honesty d) Integrity
33. If you have ____ you will be a trusted person because they will see that you are committed to you company
a) Loyalty b) Organizational Skills c) Productivity d) Respect
34. Some jobs cannot be done alone and some can be done much better and faster if you possess the quality of ____
a) Responsibility b) Respect c) Teamwork d) Cooperation
35. The three major types of ethical issues include except?
a) Communication issues b) Systematic issues
c) Corporate issues d) Individual issues
36. Ethics are moral principles and values which ____
a) Guide a firm's behavior
b) Govern the actions of an individual
c) Provide employees with rule on how to behave
d) Are legally enforceable
37. The word "Ethics" comes from
a) Ethos b) Eternity c) Elementary d) Essentiality
38. A ____ is a set of principle that people hold themselves to or that are held by organizations or groups
a) Code of ethics b) Code of honor c) Ethical perspective d) Persuasive tactic

39. _____ are beliefs and attitudes we have that can actually conflict with our ethical decision
a) Morals b) Principles c) Virtues d) Values
40. Intellectual property rights are an important aspect in international trade. Which of the following is covered under IPR?
a) Patents b) Copyrights c) Trademarks d) All of these
41. Which of the following is not the infringement of copyright?
a) Copy the software to another computer by educational institution
b) Copy the software to another computer by company
c) Copy the software to another computer
d) To make backup copies
42. Patent can be granted for _____?
a) Process b) Ideas
c) Both product and process d) Product
43. Government may order the non advertisement of any patent application in the case of invention related to _____
a) Information technology b) Food
c) Drug d) Defense technology
44. Patent for the _____ is prohibited
a) Traditional knowledge b) Arms
c) Atomic energy d) Junk food
45. A design can be also registered under _____
a) Trademark Act b) Patent Act
c) Geographical Indication Act d) Copyright Act
46. IPR Complete specification gives _____
a) Full description of claim only
b) Full description of royalty
c) Full description of invention
d) Full description of application
47. Is the first owner of a Book's copyright?
a) Author b) Publisher c) Translator d) Printing press owner
48. Copyright is a _____
a) Creation by person of efforts, intellectual and capital
b) Creation by person of the efforts only
c) Creation by negligence of a person
d) Creation by the mental status of a person
49. Globalization has improved the living structure of which of the following?
a) All the people
b) People living in developing countries
c) People living in developed countries
d) None of these
50. Which sector experienced the least profit due to the effect of Globalization?
a) Industrial sector b) Service sector c) Agriculture Sector d) Secondary sector

CBCS SCHEME

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BEC501

Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

Technological Innovation and Management

Entrepreneurship

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	Explain the different roles played by Managers.	10	L1	CO1
	b.	Describe the managerial skills required using skill-mix diagram.	10	L1	CO1
OR					
Q.2	a.	Explain the various steps involved in planning.	10	L2	CO1
	b.	Explain all the steps in rational decision making with a neat diagram.	10	L2	CO1
Module – 2					
Q.3	a.	Define organization. Briefly explain the principles of organizing.	10	L2	CO2
	b.	What is recruitment? Explain the steps in the selection process.	10	L2	CO2
OR					
Q.4	a.	Explain Maslow's need hierarchy theory with a neat diagram along with examples.	10	L2	CO2
	b.	Discuss Autocratic, Democratic and Free-rein leadership styles.	10	L2	CO2
Module – 3					
Q.5	a.	Define Social Audit. Explain the benefits and limitations of social audit.	10	L2	CO3
	b.	Explain the different views on social responsibility of business.	10	L2	CO3
OR					
Q.6	a.	Explain different types of entrepreneurs by defining an entrepreneur.	10	L2	CO3
	b.	Explain entrepreneurial development cycle.	10	L2	CO3
Module – 4					
Q.7	a.	Explain the different Government policy and development of the small scale sector in India.	10	L2	CO4
	b.	Explain the problems for small scale industries.	10	L2	CO4
OR					
Q.8	a.	Explain the identification of business opportunities in India.	10	L2	CO4
	b.	Explain in detail the project feasibilities.	10	L2	CO4
Module – 5					
Q.9	a.	What are the reasons for failure of some business plans?	10	L2	CO4
	b.	Explain the Government schemes for funding business.	10	L2	CO4
OR					
Q.10	a.	Explain the challenges and difficulties in starting an enterprise.	10	L2	CO4
	b.	Describe the limitations and differences of PERT and CPM.	10	L2	CO4

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BEC502

Fifth Semester B.E/B.Tech. Degree Examination, Dec.2024/Jan.2025 Digital Signal Processing

Time: 3 hrs.

Max. Marks:100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

Module – 1			M	L	C
1	a.	List and discuss different discrete time signals.	7	L2	CO1
	b.	Explain the steps of converting analog to digital signal in terms of frequencies.	7	L2	CO1
	c.	Discuss the advantages and limitations of Digital Signal Processing (DSP).	6	L2	CO1
OR					
2	a.	With an example, explain how to verify any signal is periodic or Not.	6	L2	CO1
	b.	Derive the equation for output of a LTI system and list the steps of convolution.	8	L3	CO2
	c.	Write a program to generate : i) Circuit step sequence ii) Sinusoidal sequence.	6	L3	CO2
Module – 2					
3	a.	Describe the properties of Z – transformation.	7	L3	CO2
	b.	Show that Discrete Fourier Transform (DFT) is a Linear Transformation.	7	L3	CO2
	c.	Compute the A-point DFT of $x(n) = \{1, 1, 0, 0\}$.	6	L3	CO2
OR					
4	a.	Compute the N-point DFT of, $x(n) = e^{j\omega n}$.	6	L3	CO2
	b.	State and prove symmetry property of DFT for real valued sequence.	6	L3	CO2
	c.	Compute circular convolution of sequences : $x_1(n) = \{2, 1, 2, 1\}$ and $x_2(n) = \{1, 2, 3, 4\}$.	8	L3	CO2
Module – 3					
5	a.	State and prove circular time shift property of DFT.	6	L3	CO2
	b.	Compare DFT and FFT with examples.	6	L2	CO3
	c.	Compute Radix – 2 DIT FFT of the following – sequence, $x(n) = n + 1$, for $0 \leq n \leq 7$.	8	L3	CO3
OR					
6	a.	State and prove Parseval's theorem for – DFT's.	6	L3	CO2
	b.	Explain overlap – save method used for the convolution of long input sequences.	6	L2	CO3
	c.	Develop an algorithm for Radix – 2 FFT without using built in function.	8	L3	CO3
1 of 2					

Module – 4

7	a.	Obtain the frequency response expression for the symmetric linear phase FIR filter.	8	L3	CO4
	b.	Compare different widows used to design FIR filters.	6	L2	CO4
	c.	Design an FIR filter using hamming window for $N = 7$. The desired frequency response is given by $H_d(\omega) = \begin{cases} e^{-j3\omega} & \omega \leq \frac{3\pi}{4} \\ 0, & \frac{3\pi}{4} < \omega \leq \pi \end{cases}$	6	L3	CO4

OR

8	a.	Discuss the characteristics of practical frequency selective filters.	6	L3	CO4
	b.	Explain the steps of designing linear phase FIR high pass filter.	8	L2	CO4
	c.	Realize the system function of following FIR filter in cascade form. $H(z) = 1 - 2z^{-1} + \frac{1}{2}z^{-2} + \frac{1}{2}z^{-3} - \frac{1}{2}z^{-4}$.	6	L3	CO4

Module – 5

9	a.	Explain the design procedure of analog Butter worth lowpass prototype – filter?	8	L3	CO5
	b.	Construct the system function in S – domain for $N = A$.	6	L3	CO5
	c.	Realize direct form – II for the IIR filter represented by $y(n) - \frac{1}{4}y(n-1) + \frac{1}{8}y(n-2) = x(n) + \frac{1}{2}x(n-2)$.	6	L3	CO5

OR

10	a.	Design the digital IIR filter for following details. -3dB gain at 0.5π rads and the stop band automation of 15dB at 0.75π rads. Assume $T_s = 15$.	8	L3	CO5
	b.	Explain the significance of: i) Prewarping ii) Bilinear transformation.	6	L2	CO5
	c.	Obtain the direct form-I realization of following IIR filter : $H(z) = \frac{1 + 0.4z^{-1}}{1 - 0.5z^{-1} + 0.06z^{-2}}$	6	L3	CO5

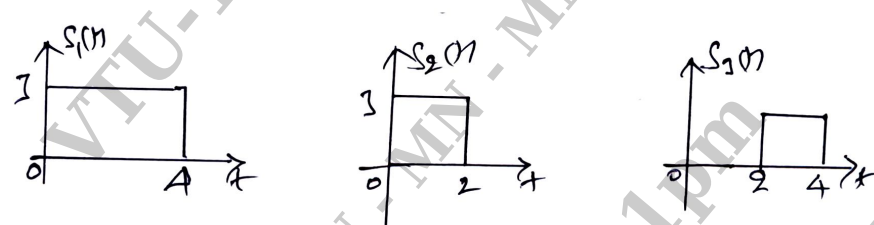
Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

Digital Communication

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1				M	L	C
Q.1	a.	Explain Hilbert transform and its properties.		6	L2	CO1
	b.	Describe the canonical representation of bandpass signal.		7	L2	CO1
	c.	Describe the correlation receiver with neat diagram.		7	L2	CO1
OR						
Q.2	a.	Apply gram Schmidt orthogonalization procedure find the set of orthonormal basis function to represent the signals $S_1(t)$, $S_2(t)$ and $S_3(t)$ as shown in Fig.Q2(a). Also express each of these figures in terms of set of basis function.		10	L3	CO1
		 Fig.Q2(a)				
	b.	Derive the equation for converting continuous AWGN channel into a vector channel.		10	L2	CO1
Module – 2						
Q.3	a.	Describe with a neat diagram, the generation and detection of BPSK signal.		8	L2	CO2
	b.	Define bandwidth efficiency. Tabulate the comment on the bandwidth efficiency of M-ary PSK signal.		8	L2	CO2
	c.	Encode the binary sequence using DPSK 11011011. Assume reference bit as 1.		4	L2	CO2
OR						
Q.4	a.	Derive the expression for probability of error of QPSK signal.		8	L2	CO2
	b.	Discuss the non-coherent detection of BFSK signal.		8	L2	CO2
	c.	Calculate the average power required for a DPSK signal operation at a data rate of 1000 bit/sec, over a band-pass channel having a bandwidth of 3000 Hz, $\frac{N_0}{2} = 10^{-10}$ W/Hz probability of error $P_e = 10^{-5}$.		4	L3	CO2
Module – 3						
Q.5	a.	Define entropy and summarize its properties.		6	L2	CO3
	b.	A source has five symbols $S = \{S_1, S_2, S_3, S_4, S_5\}$ with probabilities $P = \{0.4, 0.2, 0.2, 0.1, 0.1\}$ respectively. compute the source code using Huffman binary coding. Also find the average length and entropy.		8	L3	CO3
	c.	Briefly discuss instantaneous code with an example.		6	L2	CO3
OR						
Q.6	a.	Derive the expression for mutual information and summarize its properties.		10	L2	CO3
	b.	Derive the expression for the channel capacity of binary symmetric channel.		10	L3	CO3

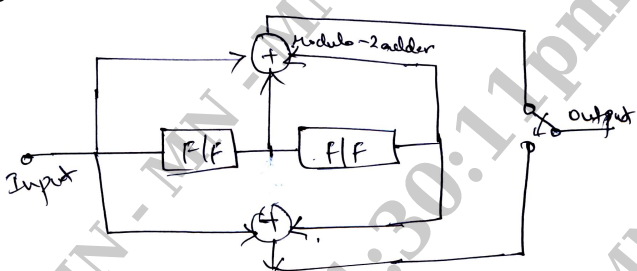
Module – 4

Q.7	a.	Indicate the advantages and disadvantages of error control coding. Also differentiate between block code and convolution code.	8	L2	CO4
	b.	If 'C' is a valid code vector then show that $CH^T = 0$ where H is parity check matrix of code.	5	L2	CO4
	c.	Design an encoder for the (7, 4) binary cyclic code generated by : $g(x) = 1 + x + x^3$ for the message vector [1001].	7	L3	CO4

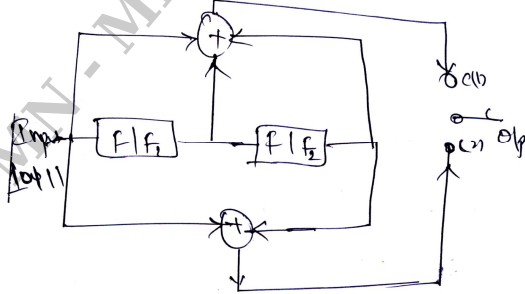
OR

Q.8	a.	Describe the block diagram of generator and parity check matrix with equation. Also write the syndrome equation and list its properties.	10	L2	CO4
	b.	A (7, 4) Linear block code has : $P = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$ i) All possible code vector ii) Determine the Hamming weight of each code word iii) If the received vector is [1100010]. Determine its syndrome correct the codeword.	10	L3	CO4

Module – 5

Q.9	a.	For a given convolutional encoder shown in Fig.Q9(a), with D = 10011. Compute output sequence using transform domain approach. Also draw the code free diagram.	10	L3	CO5
		 Fig.Q9(a)	10	L3	CO5
	b.	Describe the recursive systematic convolutional code encoder with an example.	10	L3	CO5

OR

Q.10	a.	A convolution encoder has two flip-flop with two states, three modulo – 2 adders and an output multiplexer. The generator sequences of the encoder. $g^{(1)} = (1, 0, 1)$, $g^{(2)} = (1, 1, 0)$, $g^{(3)} = (1, 1, 1)$. i) Generator matrix [G] ii) Draw the encoder block diagram iii) Calculate the codeword for the message input vector 11101.	10	L3	CO5
	b.	For a given convolution encoder shown in Fig.Q10(b). Build state table, state transaction table, sketch diagram and describe the Trellis diagram for the input message vector (10111).	10	L3	CO5
		 Fig.Q10(b)			

CBCS SCHEME

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BEC515D

Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Satellite and Optical Communication

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	Explain the Kepler laws of planetary motion. Also derive the expression for orbital period.	10	L2	CO1
	b.	A satellite is orbiting earth in a uniform circular orbit at a height of 630 km from the surface of earth. Assuming the radius of earth and its mass to be 6370 km and 5.98×10^{24} kg respectively. Determine the velocity of the satellite. (Take gravitational const $G = 6.67 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$).	10	L3	CO1
OR					
Q.2	a.	The apogee and perigee distance of satellite orbiting in an elliptical orbit are respectively, 45000 km and 7000 km. Determine the followings: i) Semi-major axis of the elliptical orbit. ii) Orbit eccentricity iii) Distance between the center of earth and the center of elliptical orbit.	10	L3	CO1
	b.	Explain briefly any six orbital parameters required to determine a satellite orbit.	10	L2	CO1
Module – 2					
Q.3	a.	Explain the satellite subsystems.	10	L2	CO2
	b.	Explain the solar energy driven power supply system of a satellite.	10	L2	CO2
OR					
Q.4	a.	Describe the telemetry, telecommand and tracking control monitoring system of a communication satellite.	10	L2	CO2
	b.	Explain with block schematic arrangement of a generalized earth's station.	10	L2	CO2
Module – 3					
Q.5	a.	What is transponder? Explain the various types of transponders.	10	L2	CO3
	b.	List the advantages and disadvantages of satellites with respect to terrestrial networks.	10	L1	CO3
OR					
Q.6	a.	Explain with a neat diagram satellite point-to-point telephonic network.	10	L2	CO3
	b.	Explain with a neat diagram satellite – cable TV.	10	L2	CO3
1 of 2					

Module – 4

Q.7	a.	Explain the mode theory as applied to circular wavelength (wave guides) in optical fibers.	10	L2	CO4
	b.	Describe the operational difference between single-mode and multimode fibers in terms of bandwidth and attenuation.	10	L2	CO4

OR

Q.8	a.	What is modal delay and how does it contribute to modal dispersion in multimode fibers?	10	L2	CO4
	b.	Define material dispersion and explain how it arises in optical fibers.	10	L2	CO4

Module – 5

Q.9	a.	Explain the principle operation of LED's.	10	L2	CO5
	b.	Discuss the characteristics of the optical detectors.	10	L2	CO5

OR

Q.10	a.	Explain the principle operation of WDM standards.	12	L2	CO5
	b.	Explain the isolators and circulators.	8	L2	CO5

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Question Paper Version : A

Fifth Semester B.E./B.Tech Degree Examination, Dec.2024/Jan.2025
Environmental Studies

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the **fifty** questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

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1. Which of the following is a Producer?
 a) Animals b) Human beings c) Plants d) Fish
 2. An ecosystem consists of
 a) Biotic species b) Biotic species and physical factors
 c) Only insects d) Only materials
 3. The term Ecology was coined by
 a) E.P. Odium b) A. G. Tansley c) Albert Einstein d) A.G. Haeckal
 4. Which year did the concept of sustainability first appear?
 a) 1992 b) 1978 c) 1980 d) 1987
 5. According to United Nations, the following are the significant issues of social progress.
 a) Instruction b) Public Health c) Living standards d) All of these
 6. The number of SDGS are
 a) 10 b) 15 c) 17 d) 16
 7. Which one of the following is not comes under SDG's?
 a) Zero Hunger b) Political activity c) Education d) Gender quality
 8. Which of the following has the largest population in food chain?
 a) Producers b) 1st degree consumers
 c) 2nd degree consumers d) Decomposers
 9. U.N era of scheduling for sustainable growth is from
 a) 2002 - 11 b) 2003 - 12 c) 2004 - 13 d) 2005 - 14

10. Which of the following is an example of Artificial Ecosystem?
a) Forest b) River c) Aquarium d) Lake
11. Solar Energy is an example of
a) Renewable b) Continuous c) Non - renewable d) None of these
12. Hydrogen energy can be tapped through
a) Heat pumps b) Fuel cells c) Petroleum d) Gassifiers
13. OTEC is an energy technology that converts
a) Energy in large tides of ocean to generate electricity
b) Energy in ocean waves to generate electricity
c) Energy in ocean due to thermal gradient to generate electricity
d) Energy in the fast moving ocean currents to generate electricity.
14. Which of the following source of energy is less eco – friendly?
a) Biogas b) Wind c) Solar d) Nuclear
15. O₃ Represents
a) Oxygen b) Planets c) Ozone d) Green house gases
16. Which of the following is an example of Water Borne Diseases
a) Hepatitis - A b) Hepatitis - B c) Hepatitis - C d) Hepatitis - H
17. Identify the renewable sourced of energy from the following :
a) Coal b) Wind power c) Uranium d) Oil
18. Which of the following is biotic?
a) Forest b) Iron c) Water d) None
19. World Environment Day celebrated on
a) 6th June b) 5th July c) 5th June d) 5th Jan
20. Which of the following is conventional source of Energy?
a) Wind Energy b) Solar Energy c) Bio gas d) None of these
21. Temporary hardness of water is due to
a) Chloride hardness b) Manganese hardness
c) Calcium hardness d) Carbonate hardness
22. The green house gas is
a) N₂O b) CH₄ c) CO₂ d) All of these
23. Which is not a natural source of pollution?
a) Volcanoes b) Forest - Fire c) Coal - fire d) Dust storms
24. Air pollution from automobiles can be controlled by providing
a) Wet collector b) Scrubbers c) Catalytic converter d) All of these
25. Normal sound level of quiet speech is
a) 120 dB b) 90 dB c) 140 dB d) 50 dB

26. What is the permissible range of p^H for drinking water as per the Indian standard?
 a) 6 to 9 b) 6.5 to 7.5 c) 6 to 8.5 d) 6.5 to 8.5
27. Which of the following is a major cause of soil pollution?
 a) Accidents involving the vehicles that are transporting waste material.
 b) Pesticides and chemical fertilizers from agricultural lands
 c) Improper solid waste disposal d) All of these
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 a) Kytoto protocol b) Basal protocol c) Montreal protocol d) Viema protocol
37. The Ozone layer is locate upto _____ km above the earth surface.
 a) 1000 b) 50 c) 80 d) 100
38. Acid rain effects on
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39. High – Concentration of fluoride content in drinking water causes.
 a) Dental flurosis b) Bone brittling c) Dental caries d) Plague
40. Desired concentration of fluoride content in drinking water in mg/ℓ is
 a) 0.7 – 1.2 b) 0.5 – 2 c) 2 – 5 d) 1 – 1.75

41. The First United National conference as Human Environment was held at Stickholm in
a) December 1972 b) June 1972 c) June 1974 d) June 1992
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49. The Organic material of the solid waste will decompose.
a) By the flow of water b) By the soil particles
c) By the action of micro organisms d) B Oxidation
50. Which of the following element make e – waste hazardous in nature?
a) Lead b) Glass c) Plastic d) Iron

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Question Paper Version : B

Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025
Environmental Studies

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the **fifty** questions, each question carries one mark.
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a) Energy in large tides of ocean to generate electricity
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c) Energy in ocean due to thermal gradient to generate electricity
d) Energy in the fast moving ocean currents to generate electricity.
24. Which of the following source of energy is less eco – friendly?
a) Biogas b) Wind c) Solar d) Nuclear

25. O₃ Represents
a) Oxygen b) Planets c) Ozone d) Green house gases
26. Which of the following is an example of Water Borne Diseases
a) Hepatitis - A b) Hepatitis - B c) Hepatitis - C d) Hepatitis - H
27. Identify the renewable sourced of energy from the following :
a) Coal b) Wind power c) Uranium d) Oil
28. Which of the following is biotic?
a) Forest b) Iron c) Water d) None
29. World Environment Day celebrated on
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30. Which of the following is conventional source of Energy?
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33. The term Ecology was coined by
a) E.P. Odiyum b) A. G. Tansley c) Albert Einstein d) A.G. Haeckal
34. Which year did the concept of sustainability first appear?
a) 1992 b) 1978 c) 1980 d) 1987
35. According to United Nations, the following are the significant issues of social progress.
a) Instruction b) Public Health c) Living standards d) All of these
36. The number of SDGS are
a) 10 b) 15 c) 17 d) 16
37. Which one of the following is not comes under SDG's?
a) Zero Hunger b) Political activity c) Education d) Gender quality
38. Which of the following has the largest population in food chain?
a) Producers b) 1st degree consumers
c) 2nd degree consumers d) Decomposers
39. U.N era of scheduling for sustainable growth is from
a) 2002 - 11 b) 2003 - 12 c) 2004 - 13 d) 2005 - 14
40. Which of the following is an example of Artificial Ecosystem?
a) Forest b) River c) Aquarium d) Lake

41. Temporary hardness of water is due to
a) Chloride hardness b) Manganese hardness
c) Calcium hardness d) Carbonate hardness
42. The green house gas is
a) N_2O b) CH_4 c) CO_2 d) All of these
43. Which is not a natural source of pollution?
a) Volcanoes b) Forest - Fire c) Coal - fire d) Dust storms
44. Air pollution from automobiles can be controlled by providing
a) Wet collector b) Scrubbers c) Catalytic converter d) All of these
45. Normal sound level of quiet speech is
a) 120 dB b) 90 dB c) 140 dB d) 50 dB
46. What is the permissible range of p^H for drinking water as per the Indian standard?
a) 6 to 9 b) 6.5 to 7.5 c) 6 to 8.5 d) 6.5 to 8.5
47. Which of the following is a major cause of soil pollution?
a) Accidents involving the vehicles that are transporting waste material.
b) Pesticides and chemical fertilizers from agricultural lands
c) Improper solid waste disposal d) All of these
48. What does E – Waste stand for?
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a) The placement on the land b) The lack of waste
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50. What is major cause of air pollution?
a) Fossil fuel burning b) Industry emission
c) Wild fire d) All of these

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Question Paper Version : C

Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025
Environmental Studies

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

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Question Paper Version : D

Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025
Environmental Studies

Time: 1 hr.]

[Max. Marks: 50

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Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025
Research Methodology and IPR

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.

2. M : Marks , L: Bloom's level , C: Course outcomes.

Module – 1			M	L	C
Q.1	a.	Identify the meaning of Research and brief out the objective and motivation in engineering research.	10	L1	CO1
	b.	Explain brief about research cycle and verify with the research flow diagram.	10	L1	CO1
OR					
Q.2	a.	Identify the types of engineering research and briefly explain them.	10	L1	CO1
	b.	Explain about the different types of research misconduct.	10	L1	CO1
Module – 2					
Q.3	a.	Explain about the importance of literature review and technical reading.	10	L2	CO2
	b.	Mention the various benefits of bibliographic databases.	10	L1	CO2
OR					
Q.4	a.	Identify the impact of technical reaction and brief about it.	10	L1	CO2
	b.	Enumerate the impact of title and keywords on citation with example.	10	L2	CO2
Module – 3					
Q.5	a.	Define Intellectual properties and explain about its types.	10	L1	CO3
	b.	Explain about the key aspect of patent law.	10	L2	CO3
OR					
Q.6	a.	Explain about the assessment of novelty.	10	L1	CO3
	b.	Brief about the patent procedure in India.	10	L1	CO4
Module – 4					
Q.7	a.	Mention and brief about the justification for copyright law.	10	L2	CO4
	b.	Explain about the basic concepts of under lying copyright law.	10	L1	CO4
OR					
Q.8	a.	Brief about the various representations of sound recordings.	10	L2	CO5
	b.	Explain about TRIPS agreement in detail.	10	L1	CO5

Module – 5					
Q.9	a.	Explain about the justification of protection designs.	10	L2	CO5
	b.	Brief about the excluded subjected matter in the context of design protection.	10	L1	CO5
OR					
Q.10	a.	What are the rights of the owner of designs? Explain.	10	L1	CO5
	b.	Brief about the Assignment of Design Rights.	10	L1	CO5
